



CEO tenure and labor investment efficiency

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ABSTRACT

We examine the impact of CEO tenure on corporate labor investment efficiency. While some studies show that CEO entrenchment increases in tenure (e.g., Hermalin and Weisbach, 1998), others argue that managerial expertise builds over time in the office (e.g., Graf-Vlachy et al., 2020). Using a large sample of U.S. firms from 1992 to 2018, we find that longer-tenured CEOs are associated with more efficient labor investment. This finding is robust to alternative measures of labor investment inefficiency, subsample analyses, and various model specifications. Further analysis shows that longer-tenured CEOs mitigate both under- and over-investment in labor problems characterized by under-hiring and under-firing, respectively. The positive effect of CEO tenure on labor investment efficiency is stronger in human capital-intensive firms and those with greater operational uncertainty. We conclude that advanced managerial expertise enables longer-tenured CEOs to invest in labor more accurately.

1. Introduction

Employment decisions are important because labor is a critical factor of production used by all firms. Labor expenditures are material, and labor investments are economically significant (Jung et al., 2014; Traini et al., 2024).¹ Inefficient labor investment can negatively affect a firm's future profitability, growth, and value (Merz and Yashiv, 2007; Jung et al., 2014; Lai et al., 2021). Since human capital is one of the key factors for firms' competitiveness and success (Pfeffer, 1996), CEOs are often involved in hiring decisions (Khedmati et al., 2020). Studies have documented profound empirical evidence that the CEO of a firm plays a critical role in the firm's labor investment and affects labor investment efficiency (Pan et al., 2016; Khedmati et al., 2020; Lai et al., 2021).² CEO tenure, as a vital CEO characteristic, affects many firm policies and outcomes, including labor investment.³ Pan et al. (2016) document a positive association between CEO tenure and the labor growth rate. However, the substantial employment growth rate over a CEO's tenure can result from successful expansion or empire-building. Understanding the true impact of CEO tenure on corporate labor investment efficiency sheds light on the ongoing debate on CEO term limits.

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¹ Jung et al. (2014) note that labor costs represent about two-thirds of economy-wide value added. The 2018–2020 annual survey of Manufactures by Census Bureau reports that payroll totals \$693.18 billion for the manufacturing sector in 2020, while capital expenditures are only \$332 billion. (<https://data.census.gov/table?q=AM1831BASIC&tid=ASMAREA2017.AM1831BASIC01&hidePreview=true>).

² For example, overconfident CEOs (Lai et al., 2021) and CEOs having stronger ties with independent directors (Khedmati et al., 2020) are associated with inefficient labor investment.

³ See detailed discussions in the section of literature review (Section 2.1).

Suboptimal labor investment results from either over- or under-investment in labor. Over-investment in labor refers to over-hiring and/or retaining employees in under-performing projects (Williamson, 1963; Ben-Nasr and Alshwer, 2016; Jung et al., 2014). Managers may increase the labor force beyond the optimal level to gain more security, power, prestige, and professional attainment (Williamson, 1963). Over-investing in labor imposes an extra financial burden on a firm. Under-investment in labor refers to employment practices of under-hiring and/or over-firing employees important to executing a profitable project (Ghaly et al., 2015; Jung et al., 2014). Further, managers' preference for a quiet life and reluctance to expand investment in labor may also lead to under-investment in labor (Bertrand and Mullainathan, 2003). A workforce shortage can force a firm to give up profitable projects, consequently hindering its growth and value creation. Expanding or shrinking the labor force beyond what is needed can bring financial problems to a firm (Merz and Yashiv, 2007; Lai et al., 2021).

Labor investment efficiency may vary with CEO tenure since CEOs at various stages of their career may allocate firm resources and make strategic decisions differently due to evolving skills and priorities. A CEO's firm and industry-specific knowledge and expertise accumulate over the tenure in office (Ericsson and Towne, 2010; Graf-Vlachy et al., 2020; Leng and Pan, 2022), which enables the CEO to comprehend the firm's information better and assess its labor needs more accurately. Furthermore, a firm's operational environment is not static, and the relationships among the CEO, the board of directors, and other stakeholders evolve over time as well. As time goes, the board and the stakeholders become more familiar with the CEO's ability and management style, which eases the strength of monitoring, allowing the CEO to use his/her expertise to enhance labor investment efficiency. Moreover, a shorter-tenured CEO, subject to career concerns, may under-invest in labor due to conservatism or aggressively expand capital and labor investment to signal his/her talents to the board of directors and market participants, leading to labor investment inefficiency. These arguments indicate a positive association between CEO tenure and labor investment efficiency.

However, weakened board monitoring over a CEO's tenure could lead to managerial entrenchment (Coles et al., 2014; Graham et al., 2020). Furthermore, CEO confidence builds over tenure, altering long-tenured CEOs' risk taking incentives. Entrenched and perhaps over-confident, longer-tenured CEOs have more empire-building tendencies (Shleifer and Vishny, 1989; Jensen and Murphy, 1990), leading to over-investment in labor. Alternatively, long-tenured CEOs may prefer a "quiet life" and are disincentivized to invest in positive NPV projects or promptly adjust labor investment, which can result in either under- or over-investment in labor (McDonald and Westphal, 2003; Bertrand and Mullainathan, 2003). The reduced board monitoring may further facilitate such behavior. In other words, exacerbated agency problems in firms with a long-tenured CEO could potentially lead to inefficient labor investment. As such, whether and how CEO tenure affects labor investment efficiency remains an open empirical question.

To test the association between the two, we define labor investment inefficiency as the absolute value of the deviation from the expected level of labor investment estimated from a firm's economic fundamentals, including growth opportunities, profitability, size, liquidity, debt usage, and loss occurrence (Pinnuck and Lillis, 2007; Jung et al., 2014; Ben-Nasr and Alshwer, 2016; Ghaly et al., 2020). The higher the deviation, the greater the labor investment inefficiency. CEO tenure is the number of years that an individual serves as the CEO of a firm (Simsek, 2007; Souder et al., 2012; Luo et al., 2014; Pan et al., 2016). Using a sample of 21,603 firm-year observations of U.S. firms from 1992 to 2018, we find a negative and significant association between CEO tenure and labor investment inefficiency. Our baseline result shows that a one-standard-deviation increase in CEO tenure is associated with a decrease in labor investment inefficiency by 3.04 % of its standard deviation. The standardized coefficient analysis reveals that the impact of CEO tenure on labor investment efficiency is either greater than or equivalent to the effects of other firm characteristics. The results indicate that the impact of CEO tenure on labor investment efficiency is also economically important. Furthermore, longer-tenured CEOs help mitigate both under- and over-investments in labor from under-hiring and under-firing, respectively.

We conduct several cross-sectional analyses to investigate which mechanism/mechanisms drives/drive the positive association between CEO tenure and labor investment efficiency. High labor skills in human capital-intensive firms (Button, 1990; Hamermesh and Pfann, 1996; Ochoa, 2013; Pfann and Verspagen, 1989; Cao and Rees, 2020) and the difficulty in assessing the labor demand in firms with more operational uncertainty (Habib and Hasan, 2021) increase labor adjustment costs. If long-tenured CEOs' expertise enables them to make more efficient labor investment decisions, we expect to observe a more positive CEO tenure-labor investment efficiency association in these firms. Our results confirm our prediction. In addition, we find no evidence that corporate governance quality and CEO risk-taking incentives modify the positive effect of CEO tenure on labor investment efficiency. Furthermore, the results from mediation analysis suggest that labor investment efficiency is the mechanism through which longer-tenured CEOs improve firm operating performance. Our findings support the managerial expertise explanation for the positive association between CEO tenure and labor investment efficiency.

To check the robustness of our results, we conduct several tests. First, to address the concern that firms with long-tenured CEOs are systematically different from those with short-tenured CEOs, we use the Entropy balancing method as in Canil et al. (2019) to mitigate the differences in the distributional moments of control variables in our baseline regression between the two types of firms. Second, to alleviate the potential endogeneity concern resulting from relatively persistent CEO tenure, we use CEO turnover as an exogenous shock and regress the change in labor investment efficiency on the change in CEO tenure over various windows around the turnover. Third, we undertake an instrumental variable approach to alleviate the bias attributable to simultaneity and reverse causality. Fourth, we conduct various subsample analyses to rule out potential explanations that our findings are merely driven by other contemporaneous non-labor investments. Additionally, we control for the impact of information asymmetry to rule out the explanation that the CEO tenure-labor investment efficiency association is driven by information asymmetry. Lastly, we employ alternative definitions of labor investment inefficiency, add additional control variables to capture the effects of local CEO supply and the average CEO/director tenure of firms in the same state, and control for the effects of different incentives of newly hired CEOs and those approaching retirement. Our baseline results hold for all the tests.

This study is the first to provide empirical evidence on the direct link between CEO tenure and labor investment efficiency. We

enrich the literature in several ways. First, we contribute to the growing literature on labor investment by showing that CEO tenure is an essential determinant of efficient labor management (Jung et al., 2014; Ben-Nasr and Alshwer, 2016; Kong et al., 2018; Khedmati et al., 2020; Luo et al., 2020; Lai et al., 2021; Le and Tran, 2021; Jung et al., 2022; James et al., 2024). Second, we add to the literature on the effects of CEO tenure on corporate decisions and outcomes (Hambrick and Fukutomi, 1991; Gibbons and Murphy, 1992; Luo et al., 2014; Pan et al., 2016; Graf-Vlachy et al., 2020). More specifically, we respond to the call for research on the relation between CEO tenure and intangible assets by Pan et al. (2016).⁴ While Pan et al. (2016) find that the increasing influence of a CEO on the board over tenure leads to decreased investment quality, we identify a different pattern of the CEO cycle in labor investment by showing that the expertise accumulated over a CEO's tenure contributes to more efficient labor investment and firm performance. Finally, our findings that longer-tenured CEOs enhance corporate employment decisions provide important implications for policymakers, corporations, the board of directors, and corporate reformers. Retaining talented CEOs can improve corporate employment decisions, an important source of firm value. Shortening CEO tenure universally may not be a perfect solution to the ongoing debate on CEO term limits.

The remainder of this paper is organized as follows. Section 2 discusses the related literature and possible associations between CEO tenure and labor investment efficiency. Section 3 describes the sample, key variables, and research design. Sections 4 and 5 provide the baseline and robustness check results, respectively, and Section 6 concludes the paper.

2. Literature review and hypothesis development

2.1. CEO tenure

A learning curve exists everywhere, and a long tenure in a specific position helps individuals improve their knowledge, decision-making efficiency, and performance. Like any other professionals, CEOs gain knowledge and experience and increase competency gradually on the job. Serving in the CEO position for a long time facilitates CEOs to know their firms and the corresponding industries well. CEOs' firm-specific knowledge and skills advance in tenure (Hambrick and Fukutomi, 1991; Leng and Pan, 2022). The increased role-specific knowledge enhances CEOs' comprehension of job responsibilities in general and tasks in specific with greater differentiation and nuance (Ericsson and Towne, 2010; Graf-Vlachy et al., 2020), which enables the CEOs to make better estimates of product demand, build good relationships with business partners and various stakeholders, project their firms' future growth and resource needs more accurately, and allocate firm resources more efficiently.

On the other hand, serving in the office for a long time may disincentivize CEOs from being open-minded, subjecting them to more "status quo" (McDonald and Westphal, 2003). Hambrick and Fukutomi (1991) argue that while firm-specific knowledge and CEO expertise advance in tenure, longer-tenured CEOs can become more powerful and less incentivized to adapt to a dynamic environment. In contrast to longer-tenured CEOs, shorter-tenured CEOs are subject to more career concerns (Fama, 1980; Hermalin and Weisbach, 1998; Holmström, 1982; Gibbons and Murphy, 1992). CEO risk-taking incentives evolve in CEO tenure as the career concern decreases as time goes by (Hambrick and Fukutomi, 1991; Levinthal and March, 1993; Luo et al., 2014). Also, both the board of directors and the market are uncertain about a novice CEO's ability and may exert more monitoring on short-tenured CEOs. However, as uncertainty can be resolved through periodic firm performance evaluations, the need to monitor and the vigilance of monitoring decrease in CEO tenure due to historical good performance.⁵ The board of directors and market participants may also intentionally or unintentionally relax the extent of monitoring due to possible reduced agency problems.⁶ Additionally, Graf-Vlachy et al. (2020) show that accumulated expertise over CEO tenure promotes more complex thinking. However, the enhanced cognitive complexity can have either positive (Streufert et al., 1968; Wong et al., 2011; Maak et al., 2016; McNamara et al., 2002) or negative implications (Downey and Slocum, 1982; Moore and Tenbrunsel, 2014; Finkelstein et al., 2009).

Given the dynamic features of CEOs during their tenure, changing relationships with the board of directors and market participants, and ever-changing internal and external operating environment, firms' activities vary over their CEOs' time in office. Studies have shown that CEO tenure affects various corporate decisions and outcomes either positively or negatively. For example, firms with long-tenured CEOs have lower cost of debt (Owusu et al., 2022), greater invention in stable technology contexts (Wu et al., 2005), less earnings management and reduced audit fees (Ali and Zhang, 2015), and strengthened firm-employee relationships (Luo et al., 2014). Conversely, researchers find that a long CEO tenure is associated with lower quality of capital investments (Pan et al., 2016), lower bond rating quality (Zhang and Schloetzer, 2021), lower likelihood of performance-related dismissals (Dikolli et al., 2014),⁷ weaker pay-performance sensitivity (Hill and Phan, 1991), lower board independence (Graham et al., 2020), reduced corporate social responsibility performance (Chen et al., 2019), and declined firm value (Brochet et al., 2021). Additionally, Hou et al. (2017) find

⁴ Labor is one of the most important intangible assets (Barth et al., 2001; Gu and Wang, 2005).

⁵ Dikolli et al. (2014) document that the likelihood of performance-related CEO dismissal declines in CEO tenure, indicating that periodic performance reports can resolve uncertainty of executives' ability, and hence lower firm owners' demand for intense CEO monitoring. Similarly, Zhang and Schloetzer (2021) argue that bond rating agencies learn about the management as management tenure advances, which could reduce the rating agencies' caution and efforts in management assessment.

⁶ More severe agency problems can occur as CEO tenure advances due to well-established relationships with the board and other market participants (Coles et al., 2014; Graham et al., 2020).

⁷ Differently, Brookman and Thistle (2009) find that the risk of termination increases in tenure and only slightly declines for CEOs with a very long tenure.

declining/increasing benefits to shareholders from CEO performance-based/non-performance-based pay as CEO tenure increases. Hambrick and Fukutomi (1991) propose that CEO tenure can be segmented into different “seasons” and the effects of CEO tenure on CEO behavior and a firm vary over these seasons.

2.2. Labor investment efficiency

Labor is a crucial input vital to the success of all organizations. Labor expenditures are material for a broad spectrum of firms (Sualihu et al., 2021a; Traini et al., 2024). Highly regulated labor markets and limited availability of skilled labor in the U.S. make it costly for firms to undertake employment adjustments (Banker et al., 2018; He et al., 2020). Furthermore, labor investments are not under the complete control of managers since employees, to some extent, have the option to determine where to work and how long to work for an employer (Donangelo, 2014; Sualihu et al., 2021b). Labor investments are the results of bi-direction selections by employers and employees (Khedmati et al., 2020).⁸ These two characteristics of labor investment may increase labor costs and reduce managerial discretion over employment decisions. Sualihu et al. (2021a) posit that labor expenditure is often prone to inefficiencies. Inefficient labor investments can deter firm growth and lead to poor firm performance and low returns to investors (Merz and Yashiv, 2007; Jung et al., 2014; Ghaly et al., 2020; Lai et al., 2021).

The vast literature on investment efficiency suggests that information asymmetry between firms and external investors results in suboptimal labor investment. Transparent firms, such as those with high accounting quality (Jung et al., 2014), high stock price informativeness (Ben-Nasr and Alshwer, 2016), more analysts’ coverage (Lee and Mo, 2020), less analysts’ forecast error (Sualihu et al., 2021b), and high stock liquidity (Ee et al., 2022), are associated with enhanced labor investment efficiency. Suboptimal labor investments may also reflect managerial opportunistic behaviors. Firms that lack effective monitoring, such as those with fewer long-term institutional investors (Ghaly et al., 2020) and stronger CEO-director ties (Khedmati et al., 2020), have less efficient labor investment. Using international data, Le and Tran (2021) show that board reform increases labor investment efficiency by mitigating managerial moral hazard.

Furthermore, operational uncertainty/risk reduces labor investment efficiency as accurately forecasting labor demand in the presence of high uncertainty/risk is increasingly difficult. Empirically, Habib and Hasan (2021) find that prospector-type firms have more suboptimal labor investments due to highly uncertain business strategies. Boubaker et al. (2022) document a positive association between product market competition and labor investment inefficiency. CEOs with stronger risk-taking incentives provided by stock options (Sualihu et al., 2021a) and over-confident CEOs (Lai et al., 2021) are associated with less efficient labor investment. Labor investment decisions are also affected by other corporate policies, such as employee treatment (Cao and Rees, 2020), tax planning (Traini et al., 2024), and labor-friendly institutional features (Jung et al., 2022).

2.3. CEO tenure and labor investment efficiency

The mixed findings on the effects of CEO tenure on corporate decisions and outcomes indicate that the impact of CEO tenure is policy-specific. CEOs’ incentives and power levels inside their firms vary systematically over their years in office (Pan et al., 2016). Consequently, CEO tenure affects CEOs’ expertise and confidence, firm-level uncertainty, and the extent of agency problems, all of which can jointly affect labor investment decisions.

Labor investment efficiency can increase with CEO tenure because CEOs attain more skills and expertise on the job. The increased firm and industry-specific knowledge enables longer-tenured CEOs to comprehend more structured information, better assess market demand for their products and services, and more accurately forecast their labor needs. Since firms are more cautious and tend to withhold investment during uncertainty (Hanlon et al., 2017; Jacob et al., 2021), accurate assessments of future labor needs should lead to greater labor investment efficiency. Moreover, both internal and external monitors are likely to exert less intensive monitoring when they become more familiar with a CEO’s ability and management style as the CEO’s tenure advances, which may facilitate long-tenured CEOs to use their expertise and invest in labor more efficiently.⁹ On the other hand, unknown to the external labor market, though shorter-tenured CEOs have stronger incentives to deliver performance, their career concern may negatively affect their firms’ labor investment decisions. They can be too cautious to proceed with positive NPV projects, leading to under-investment in labor, or they can aggressively pursue investment opportunities to impress the board of directors and market participants, resulting in over-investment in labor. The above discussions suggest a positive effect of CEO tenure on labor investment efficiency.

However, labor investment efficiency can decrease with CEO tenure. A long CEO tenure can weaken a firm’s governance mechanisms. For example, Coles et al. (2014) find that the longer the time that a CEO is in office, the higher the percentage of board members are appointed during the CEO’s tenure, resulting in more co-opted directors who tend to be more loyal to and less likely to monitor the CEO. The relaxed monitoring from the board can exacerbate the agency problem, which facilitates long-tenured CEOs to engage in empire-building, consequently leading to over-investment in labor. Furthermore, managers sometimes sacrifice long-term firm value to meet short-term reporting goals (Graham et al., 2005; He and Tian, 2013), and they may reduce the size of the labor force

⁸ Employees can be either current or potential ones.

⁹ Adams and Ferreira (2007) propose that management-friendly boards can be optimal because boards of directors rely on CEOs to obtain information to exercise effective monitoring and CEOs tend to hide information when facing tough monitors.

due to the desire to show strong short-term performance (Graham et al., 2005; Oyer and Schaefer, 2011), which can be more easily realized under weaker monitoring.¹⁰ Additionally, boards of directors exert more monitoring and advising in the early tenure of CEOs than in their later years in office, which may help short-tenured CEOs make better labor investment decisions. Furthermore, CEOs build confidence over tenure. Over-confident CEOs tend to take on too much risk and make over-optimistic projections on their firm's future growth and prospects, thus over-investing in labor. Conversely, in the later years of their tenure, CEOs may prefer a quiet life and therefore engage in risk/effort aversion to difficult decisions (Bertrand and Mullainathan, 2003). Specifically, Bertrand and Mullainathan (2003) argue that “quiet life” incentives may induce managers to be slow in firing workers, leading to over-investment in labor or inefficient labor investment. As such, the ultimate effect of CEO tenure on labor investment efficiency remains an open empirical question.

3. Research design

3.1. Sample

We start our sample with the COMPUSTAT database from 1992 to 2018. We exclude financial (SIC 6000–6999) and utility firms (SIC 4900–4949) because these firms are highly regulated and have different asset structures. CEO data is from the EXECUCOMP database. We require that CEOs have information on age and the date of appointment as CEO. Institutional ownership is from the Thomson Financial Institutional Holdings (13F) database. The industry-level unionization rate is from the website of UNIONSTATS, which provides estimates of union membership and coverage data by industry. After merging the databases and removing observations with missing data on key variables, we have a sample of 21,603 firm-year observations.

3.2. Key variables

3.2.1. Labor investment efficiency

We follow existing studies to construct the labor investment efficiency proxy by employing a two-step approach. Specifically, we measure labor investments with net hiring (*NetHire*), constructed as the percentage change in the number of employees from the previous year. We estimate the expected level of labor investments using the model as in Pinnuck and Lillis (2007), Jung et al. (2014), and Ben-Nasr and Alshwer (2016).

$$\begin{aligned} NetHire_{it+1} = & \beta_0 + \beta_1 SGR_{it} + \beta_2 SGR_{it+1} + \beta_3 \Delta ROA_{it} + \beta_4 \Delta ROA_{it+1} \\ & + \beta_5 ROA_{it+1} + \beta_6 Return_{it+1} + \beta_7 Size_R_{it} + \beta_8 Quick_{it} + \beta_9 \Delta Quick_{it} + \beta_{10} \Delta Quick_{it+1} \\ & + \beta_{11} Leverage_{it} + \beta_{12} LossBin1_{it} + \beta_{13} LossBin2_{it} + \beta_{14} LossBin3_{it} + \beta_{15} LossBin4_{it} \\ & + \beta_{16} LossBin5_{it} + IndustryFE + \varepsilon_{it+1} \end{aligned} \quad (1)$$

We control for the demand for a firm's products and services with the sales growth rate (*SGR*). Profitable firms are likely to have a higher demand for labor. We include the change in firms' profitability (*ROA*) and its lagged value to capture the effects of changes in earnings on labor investment and the potential time lag between hiring and profitability change, respectively. *Return* is the annual stock return and is used to control for future expected growth. The effect of firm size on the level of hiring is measured with the percentile rank of the logarithm of market value (*Size_R*). Firms with higher liquidity are less likely to retrench employees due to cash flow shortages. We include quick ratio (*Quick*) and its current and lagged changes to capture this effect. We control for the impact of capital structure with *Leverage*. Firms using more debt financing have less funds available for labor investments. Lastly, we control for loss occurrence since firms incurring losses have limited cash flows and thus may cut back labor investments. *LossBin* are indicator variables for each 0.005 interval of the prior-year's ROA from 0 to −0.025 (e.g., *LossBin1* equals one if ROA in the previous year is between −0.005 and 0, *LossBin2* equals one if ROA in the previous year is between −0.01 to −0.005, etc.). We control for industry fixed effects, where the industry is defined using the Fama and French 48 industry classification. Refer to Appendix 1 for detailed definitions of these variables.

We proxy labor investment inefficiency with abnormal labor investment (*Ab_Abnormal_NetHire*), defined as the absolute value of the difference between the actual level of labor investment and the expected level estimated from Eq. (1). A higher value of *Ab_Abnormal_NetHire* suggests a larger deviation of labor investment from the expected level justified by a firm's economic fundamentals, i.e., greater labor investment inefficiency.

3.2.2. CEO tenure

CEO tenure is the number of years that an individual has served as the CEO in a hosting firm (Wu et al., 2005; Henderson et al., 2006; Walters et al., 2007; Simsek, 2007; Luo et al., 2014; Pan et al., 2016). Given that CEO tenure and labor investment inefficiency might be nonlinear, we thus use an alternative proxy, $\log(1 + \text{CEO tenure})$, defined as the natural logarithm of one plus *CEO tenure* (Onali et al., 2016). We employ Akaike and Bayes Information Criteria (denoted AIC and BIC) in our baseline regression to investigate which proxy better captures the effect of CEO tenure on labor investment efficiency (i.e., having less information loss).

¹⁰ For example, Graham et al. (2005) find that managers postpone or eliminate hiring to avoid missing earnings targets, which may result in sub-optimal employment decisions.

3.3. Empirical model

We augment the model used in the literature (Ben-Nasr and Alshwer, 2016; Ghaly et al., 2020; Jung et al., 2014; Pinnuck and Lillis, 2007) by including our variable of interest, *CEO Tenure*, to examine the impact of CEO tenure on corporate labor investment efficiency as follows:

$$\begin{aligned} Ab_Abnormal_NetHire_{it+1} = & \beta_0 + \beta_1 CEOTenure_{it} + \beta_2 CEOAge_{it} + \beta_3 MTB_{it} + \beta_4 Size_{it} + \beta_5 Quick_{it} + \beta_6 Leverage_{it} + \beta_7 DivDum_{it} \\ & + \beta_8 STD_CFO_{it} + \beta_9 STD_Sale_{it} + \beta_{10} Tangible_{it} + \beta_{11} Loss_{it} + \beta_{12} Insti_{it} + \beta_{13} STD_NetHire_{it} \\ & + \beta_{14} Labor_Intensity_{it} + \beta_{15} Union_{it} + \beta_{16} AB_Abnormal_InvestOther_{it} + IndustryFE + YearFE + \varepsilon_{it+1}, \end{aligned} \quad (2)$$

The model is estimated using OLS regression with the standard errors clustered at the firm level to generate statistical inference. We control for the impact of CEO age (*CEO Age*) to ensure that the observed CEO tenure-labor investment efficiency relation is not from the age effect. In addition, we control for the effects of growth opportunities (market-to-book value of common equity, *MTB*), firm size (the natural logarithm of a firm's market value of equity, *Size*), liquidity (quick ratio, *Quick*), financial risk (leverage, *Leverage*), payout policies (dividend payout status, *DivDum*), volatility of cash flow from operations (*STD_CFO*), volatility of revenues (*STD_Sale*), fixed assets investments (*Tangibles*), reported losses (*Loss*), institutional ownership (the proportion of common shares held by institutional owners, *Insti*), volatility of the investment in labor (*STD_NetHire*), labor intensity (the ratio of the number of employees over total assets, *Labor_Intensity*), and labor protection (the industry unionization rate, *Union*).¹¹ Lastly, we control for the impact of abnormal non-labor investments (*AB_Abnormal_Invest_Other*), estimated as the absolute value of the residuals from the model by regressing other investments on the lagged sales growth rate (*SGR*), where other investments include capital, acquisition, and R&D expenditures less the proceeds from selling property, plant, and equipment.

We determine whether a linear or nonlinear model is more appropriate for our analysis using a semi-parametric approach. Following Brochet et al. (2021), Huang and Hilary (2018), Cleveland and Devlin (1988), and Cleveland (1979), we use the locally weighted scatterplot smoothing (Lowess) procedure. Instead of requiring a specified global function to fit the model to the whole set of the data, this method uses weighted least squares to fit low-degree polynomials to subsets of the data. Hence, each function only describes the variation of a subset of the data. In the end, Lowess estimation combines the regression lines from all regressions corresponding to each subset of the data, allowing better identification of the true functional form.

We employ the Lowess regression of residuals of *Ab_Abnormal_NetHire* on *CEO Tenure*, where the residuals are estimated by regressing *Ab_Abnormal_NetHire* on all variables excluding *CEO Tenure* in Equation (2). Graph A in Fig. 1 presents the plot. The residuals do not vary with CEO tenure, indicating a linear relation between CEO tenure and labor investment efficiency in our sample. In addition, we employ Lowess regression of *Ab_Abnormal_NetHire* on *CEO Tenure* and present the figure in Graph B. This graph further confirms the linear relation between CEO tenure and labor investment inefficiency.

3.4. Sample descriptive statistics

Table 1 presents the summary statistics of the sample. The average (median) CEO tenure of the sample is 7.7 (5) years. Close to the findings of Jung et al. (2014), the mean and median values of *Ab_Abnormal_NetHire* are 0.093 and 0.058, respectively.¹² The descriptive statistics of other control variables are generally consistent with the literature, such as Biddle et al. (2009), Jung et al. (2014), and Cao and Rees (2020).

The Pearson pairwise correlation matrix (Appendix 2) shows that *CEO tenure* and *Ab_Abnormal_NetHire* have a significant and negative correlation, providing preliminary supporting evidence that longer-tenured CEOs improve labor investment efficiency. Larger, highly leveraged, more unionized, and dividend-paying firms and those with older CEOs, more fixed assets, greater institutional ownership, and higher labor intensity are associated with greater labor investment efficiency. However, unprofitable firms and those with higher liquidity, greater abnormal non-labor investments, and more volatility in cash flows, sales, and labor investment are associated with higher labor investment inefficiency.

4. Results

4.1. Baseline regression results

We run the baseline regression and report the results in Table 2. To investigate which CEO tenure proxy, *CEO tenure* or its logarithm transformation ($\log(1 + \text{CEO tenure})$), better explains the relation between CEO tenure and labor investment efficiency, we compare AIC and BIC for Equation (2) using *CEO tenure* and $\log(1 + \text{CEO tenure})$ in Models (1) and (2), respectively. AIC is smaller in Model (1) than in Model (2), and the values of BIC are similar in the two models, suggesting that the specification of Model (1) suffers less information loss. Hence, we use *CEO tenure* in the rest of the paper.

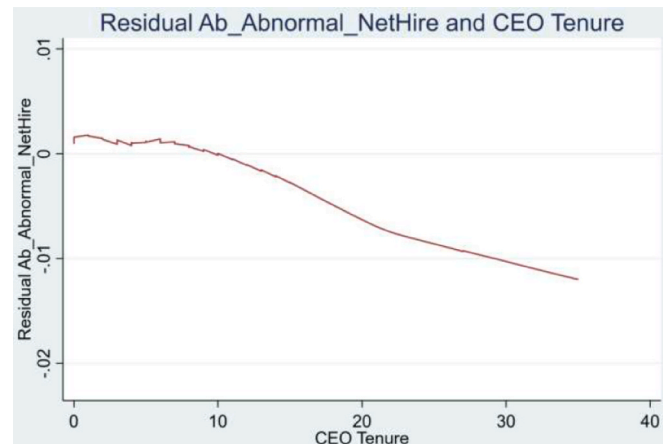
The coefficients on *CEO tenure* and $\log(1 + \text{CEO tenure})$ are negative and statistically significant at the 1 % level, suggesting that longer CEO tenure enhances labor investment efficiency. In terms of economic significance, the parameter estimate in Model (1) shows

¹¹ The database is available at <https://www.unionstats.com>. The construction of the database is detailed in Hirsch and Macpherson (2003).

¹² The mean and median values of *Ab_Abnormal_NetHire* in Jung et al. (2014) are 0.11 and 0.07, respectively.

Graph A:

This graph shows the relation between the residual of *Ab_Abnormal_NetHire* using locally weighted polynomial curve (i.e., LOWESS) and CEO tenure where the residual is obtained by regressing *Ab_Abnormal_NetHire* on the control variables except *CEO tenure*. The bandwidth is 0.6.

**Graph B:**

This graph shows the relation between *Ab_Abnormal_NetHire* using locally weighted polynomial curve (i.e., LOWESS) and CEO tenure. The bandwidth is 0.

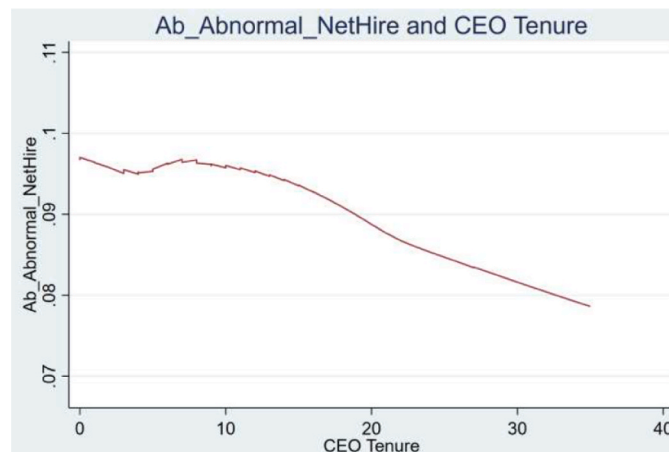


Fig. 1. Locally weighted scatterplot smoothing (Lowess) regressions. Graph A: This graph shows the relation between the residual of *Ab_Abnormal_NetHire* using locally weighted polynomial curve (i.e., LOWESS) and CEO tenure where the residual is obtained by regressing *Ab_Abnormal_NetHire* on the control variables except *CEO tenure*. The bandwidth is 0.6. Graph B: This graph shows the relation between *Ab_Abnormal_NetHire* using locally weighted polynomial curve (i.e., LOWESS) and CEO tenure. The bandwidth is 0.

Table 1

Sample statistics. This table presents descriptive statistics of the sample. The sample includes Compustat Industrial firms excluding financial (SIC 6000–6999) and utility firms (SIC 4900–4949). Firm-year observations with missing data in this study are deleted. The sample covers 21,603 firm-year observations from 1992 to 2018. Refer to Appendix 1 for detailed variable definitions. All continuous variables are winsorized at the upper and lower 1% of the sample distribution.

	<i>N</i>	<i>Mean</i>	<i>Median</i>	<i>25th percentile</i>	<i>75th percentile</i>	<i>Standard Dev.</i>
Ab_Abnormal_NetHire	21,603	0.0927	0.0576	0.0263	0.1122	0.1114
CEO tenure	21,603	7.6987	5.0000	2.0000	11.0000	7.6026
CEO age	21,603	56.0383	56.0000	51.0000	61.0000	7.3971
MTB	21,603	3.2836	2.3491	1.5196	3.8567	3.9423
SIZE (in million dollars)	21,603	6978.10	1550.29	571.35	4925.11	17605.68
Quick	21,603	1.6839	1.1995	0.7560	1.9645	1.5822
Leverage	21,603	0.2157	0.2031	0.0452	0.3316	0.1809
DivDum	21,603	0.5372	1.0000	0.0000	1.0000	0.4986
STD_CFO	21,603	0.0505	0.0391	0.0239	0.0638	0.0397
STD_Sale	21,603	0.2166	0.1534	0.0838	0.2811	0.1998
Tangibles	21,603	0.2858	0.2139	0.1009	0.4198	0.2303
Loss	21,603	0.1576	0.0000	0.0000	0.0000	0.3643
Insti	21,603	0.7358	0.7631	0.6085	0.8906	0.2075
STD_NetHire	21,603	0.1592	0.1033	0.0572	0.1860	0.1818
Labor_Intensity	21,603	0.0059	0.0034	0.0017	0.0069	0.0076
Union	21,603	0.1031	0.0752	0.0407	0.1299	0.0893
AB_Abnormal_InvestOther	21,603	0.0892	0.0725	0.0393	0.1055	0.0916

that a one-standard-deviation increase in *CEO tenure* is associated with approximately 0.08 % decrease in *Ab_Abnormal_NetHire*, equivalent to 3.04 % of the standard deviation of labor investment inefficiency in our sample firms.^{13,14} Furthermore, we standardize the coefficients in Model (1) to evaluate the economic effect of CEO tenure on labor investment inefficiency relative to those of other explanatory variables. The results show that the magnitude of the effect of CEO tenure on labor investment inefficiency is equivalent to the effects of the volatility of cash flow from operations (*STD_CFO*), reported losses (*Loss*), and labor intensity (*Labor_Intensity*), and has twice the effect of growth opportunities (*MTB*), indicating that the impact of CEO tenure is economically meaningful.¹⁵ Other firm characteristics also affect labor investment efficiency. Large, high value/growth, and high labor-intensive firms tend to make more efficient labor investment decisions. However, firms reporting losses and those having higher liquidity, higher abnormal non-labor investments, and more volatility in sales, cash flows, and labor investment are associated with less efficient labor investments.

We also investigate the explanatory power of our control variables relative to those uncontrolled in our baseline regression. To check for the potential omitted variable bias in the model specification, Oster (2017) suggests estimating the magnitude of the effect of the unobservable variables needed to outweigh that of the observable ones. As in Chintrakarn et al. (2020), we apply this methodology to our regression in Model (1), and find that only when the effect of the unobservables exceeds 1.48 times that of the observables, should our model suffer from omitted variable bias, which further strengthens our baseline regression result.¹⁶

4.2. Specific types of labor investment inefficiency

Under-hiring and over-firing employees, or simply failing to retain key employees, can lead to under-investments in labor. Investors' myopic pressure and the resultant management fear of short-term earnings and stock price drop can induce under-investment in labor (Narayanan, 1985; Stein, 1989; Von Thadden, 1995; Bushee, 1998). Over-investment in labor results from over-hiring or under-firing due to managerial empire-building (Williamson, 1963), or managerial pursuit of a quiet life and the desire to avoid making difficult decisions and costly efforts in downsizing (Bertrand and Mullainathan, 2003).¹⁷ Additionally, managers may use labor investment as a tactic to avoid dismissal for poor performance by allying with the workforce via abstaining from layoffs and wage cuts (Pagano and Volpin, 2005), leading to over-investment in labor.

Following Ben-Nasr and Alshwer (2016), Jung et al. (2014), and Ghaly et al. (2020), we conduct subsample analyses of the impact of CEO tenure on labor investment efficiency in under- and over-investment in labor firms separately. Table 3 displays the results. Model (1) shows the result from all firms with under-investment in labor (those with actual hiring less than the expected level, i.e.,

¹³ The coefficient of *CEO tenure* is -0.0004 and the standard deviation of *CEO tenure* is 2. The standard deviation of *Ab_Abnormal_NetHire* is 0.0263. A one-standard-deviation increase in *CEO tenure* is associated with a 0.08% reduction in labor investment inefficiency ($-0.0004 \times 2 = -0.0008$), corresponding to 3.04% of the standard deviation of labor investment inefficiency in our sample ($0.08\%/2.63\%$).

¹⁴ To account for the potential nonlinear relation between CEO tenure and labor investment efficiency, we include squared CEO tenure in additional analysis as a robustness check, the results continue to hold. For brevity, the results are not tabulated but available upon request.

¹⁵ The results are available upon request.

¹⁶ We employ the STATA command *psacalc* to conduct the analysis. Refer to Oster (2017) for a detailed explanation for estimating coefficient stability.

¹⁷ Williamson (1963) uses increases in employees beyond the optimal level as an example of managers' opportunistic behavior to gain more security and power.

Table 2

CEO tenure and labor investment efficiency. This table reports regression results of the effect of CEO tenure on labor investment efficiency. *Ab_Abnormal_NetHire* is the absolute value of the difference between the actual value of *NetHire* and the expected value estimated with Eq. (1). *CEO tenure* is the number of years that a CEO serves as the CEO in the hosting firm. *Log (1 + CEO tenure)* is the natural logarithm of one plus *CEO tenure*. We control industry and year-fixed effects, where the industry is defined using the Fama and French 48 industry classification. Standard errors are clustered at the firm level. Refer to Appendix 1 for detailed variable definitions. All continuous variables are winsorized at the upper and lower 1 % of the sample distribution. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

	(1)	(2)
	Ab_Abnormal_NetHire	
CEO tenure	−0.0004*** (−3.51)	
CEO age	−0.0002 (−1.34)	
Log(1 + CEO tenure)		−0.0027*** (−2.75)
Log(CEO age)		−0.0131* (−1.87)
MTB	−0.0004** (−1.96)	−0.0005** (−1.98)
Size	−0.0032*** (−5.42)	−0.0031*** (−5.32)
Quick	0.0070*** (9.86)	0.0069*** (9.76)
Leverage	0.0058 (1.12)	0.0062 (1.20)
DivDum	−0.0015 (−0.82)	−0.0015 (−0.79)
STD_CFO	0.0744*** (2.72)	0.0748*** (2.73)
STD_Sale	0.0265*** (4.63)	0.0265*** (4.62)
Tangible	0.0014 (0.24)	0.0012 (0.20)
Loss	0.0080*** (3.37)	0.0080*** (3.36)
Insti	0.0013 (0.28)	0.0021 (0.43)
STD_NetHire	0.0600*** (9.79)	0.0602*** (9.81)
Labor_Intensity	−0.4509*** (−3.35)	−0.4552*** (−3.40)
Union	0.0120 (0.66)	0.0128 (0.70)
AB_Abnormal_InvestOther	0.3791*** (24.63)	0.3794*** (24.64)
Constant	0.0726*** (2.97)	0.1155*** (3.24)
AIC	−37432.1	−37425
BIC	−36852.5	−36850.4
Observations	21,603	21,603
R-squared	0.171	0.170
Industry effects	Yes	Yes
Year fixed effects	Yes	Yes
Firm clustered standard error	Yes	Yes

Abnormal_NetHire < 0). We further focus on under-hiring (firms with positive expected net hiring) and over-firing firms (those with negative expected net hiring) in Models (2) and (3), respectively. Similarly, Model (4) shows the results of all firms with over-investment in labor (actual hiring greater than the expected level, i.e., *Abnormal_NetHire* > 0), and Models (5) and (6) display the results from over-hiring firms (those with positive expected net hiring) and under-firing firms (those with negative expected net hiring), respectively.

The coefficients on *CEO tenure* are negative and significant in Models (1) and (2), suggesting that longer-tenured CEOs alleviate under-investment in labor by reducing under-hiring. The negative coefficients on *CEO tenure* in Models (4) and (6) indicate that longer-tenured CEOs mitigate over-investment in labor by reducing under-firing. Under-investment in labor is more likely to occur during economic contractions when there are sharp declines in consumers' purchasing power (Sualihu et al., 2021a, 2021b; Jung et al., 2014) and liquidity in financial markets (Campello et al., 2010; Benmelech et al., 2015). Over-investment in labor is more likely to occur during economic expansions characterized by higher future growth prospects, during which time CEOs tend to maintain more staff for

Table 3

CEO tenure and different types of labor investment inefficiency. This table presents the results for the subsample analysis on the association between CEO tenure and labor investment inefficiency. Models (1)–(3) report the results of the subsamples of firms with under-investment in labor. Model (1) reports the results for the subsample of all under-investment firm-year observations (i.e., all sample firms with the actual labor investment smaller than the expected level of labor investment). Model (2) reports the results for the under-investment subsample of firms with positive expected level of labor investment (under-hiring). Model (3) reports the results for the under-investment subsample of firms with negative expected level of labor investment (over-firing). Models (4)–(6) report the results of the subsamples of firms with over-investment in labor. Model (4) reports the results of the subsample of all over-investment observations (i.e., all sample firms with the actual labor investment greater than the expected). Model (5) reports the results for the over-investment subsample of firms with a positive expected level of labor investment (over-hiring). Model (6) reports the results for the over-investment subsample of firms with a negative expected level of labor investment (under-firing). We control industry and year effects, where the industry is defined using the Fama and French 48 industry classification. Standard errors are clustered at the firm level. Refer to Appendix 1 for detailed variable definitions. All continuous variables are winsorized at the upper and lower 1% of the sample distribution. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	Ab_Abnormal_NetHire					
	Under-investment in labor (Abnormal_NetHire < 0)			Over-investment in labor (Abnormal_NetHire > 0)		
		Under hiring	Over firing		Over hiring	Under firing
CEO tenure	−0.0005*** (−3.06)	−0.0004*** (−2.89)	−0.0003 (−0.69)	−0.0004** (−2.43)	−0.0004 (−0.98)	−0.0004** (−2.26)
CEO age	0.0000 (0.22)	0.0001 (0.54)	−0.0003 (−0.96)	−0.0004** (−2.26)	−0.0004 (−0.99)	−0.0004** (−2.07)
MTB	−0.0006** (−2.38)	−0.0005* (−1.85)	−0.0011* (−1.71)	−0.0003 (−0.70)	−0.0012 (−1.36)	0.0002 (0.35)
Size	−0.0037*** (−5.34)	−0.0034*** (−4.35)	−0.0039** (−2.38)	−0.0028*** (−3.08)	−0.0004 (−0.22)	−0.0035*** (−3.38)
Quick	0.0078*** (9.40)	0.0078*** (8.97)	0.0090*** (4.08)	0.0058*** (5.14)	0.0072*** (3.09)	0.0049*** (3.74)
Leverage	0.0056 (0.86)	0.0068 (0.96)	0.0000 (0.00)	0.0082 (1.04)	−0.0099 (−0.61)	0.0142 (1.43)
DivDum	−0.0013 (−0.56)	−0.0018 (−0.71)	0.0011 (0.21)	−0.0006 (−0.21)	0.0028 (0.46)	−0.0022 (−0.70)
STD_CFO	0.0285 (0.87)	0.0390 (1.10)	−0.0502 (−0.64)	0.1416*** (3.21)	0.1201 (1.51)	0.1293** (2.47)
STD_Sale	0.0257*** (3.49)	0.0217*** (2.78)	0.0541*** (2.70)	0.0247*** (2.97)	0.0480** (2.16)	0.0221** (2.36)
Tangible	0.0061 (0.89)	0.0078 (1.06)	0.0003 (0.02)	−0.0045 (−0.52)	−0.0053 (−0.36)	−0.0025 (−0.24)
Loss	0.0079*** (2.79)	0.0066* (1.74)	0.0011 (0.21)	0.0092** (2.30)	0.0127** (2.24)	0.0072 (1.24)
Insti	0.0046 (0.80)	0.0042 (0.67)	0.0092 (0.74)	−0.0049 (−0.64)	−0.0179 (−1.26)	−0.0023 (−0.25)
STD_NetHire	0.0378*** (5.08)	0.0358*** (4.31)	0.0370** (2.36)	0.0854*** (9.53)	0.0389*** (2.66)	0.0981*** (9.42)
Labor_Intensity	−0.4377*** (−2.91)	−0.3656** (−2.24)	−0.5326 (−1.25)	−0.5778*** (−2.68)	−0.4676 (−1.14)	−0.5650** (−2.25)
Union	0.0054 (0.27)	0.0108 (0.53)	−0.0181 (−0.33)	0.0202 (0.77)	−0.0024 (−0.05)	0.0265 (0.87)
AB_Abnormal_InvestOther	0.4127*** (19.97)	0.4102*** (18.52)	0.4358*** (7.95)	0.3360*** (15.60)	0.3411*** (6.17)	0.3317*** (14.38)
Constant	0.0950* (1.95)	0.1175* (1.66)	0.0766 (1.26)	0.0752*** (2.68)	0.0864** (2.20)	0.0731** (2.23)
Observations	12,738	10,368	2,370	8,865	2,042	6,823
R-squared	0.175	0.183	0.170	0.177	0.216	0.180
Industry effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm clustered standard error	Yes	Yes	Yes	Yes	Yes	Yes

Table 4

Cross-sectional analyses This table reports the results of cross-sectional analysis. Panel A displays the result of the moderating effect of human capital intensity on the CEO tenure-labor investment inefficiency relation. Human capital intensity of a firm is proxied by four alternative measures, *Human capital intensity* (the ratio of R&D expenditures to total sales) in Model (1), *Human capital intensive industries* (all subcategories of the telecommunications, high-tech, and healthcare industries with the two- and three-digit SIC codes of 283, 357, 36, 384, 48 and 80) in Model (2), *Labor skill* (labor skills data from Belo et al. (2017) that uses the industry average number of employees working in occupations with a JobZones index equal to 4 or 5 as a proxy for the degree of reliance on skilled labor) in Model (3), and *Knowledge capital* (the flows of a firm's knowledge capital in the intangibles defined as in Ewens et al., 2023) in Model (4). Panel B displays the results of the moderating effect of operational uncertainty on the CEO tenure-labor investment efficiency relation using OLS regressions. Operational uncertainty is proxied by *STD_CFO* (the standard deviation of cash flows from operations over the past five years) in Model (1), *Idiosyncratic risk* (the logarithm of one plus the variance of residuals estimated from the Fama-French 3-factor model) in Model (2), and *Total risk* (the logarithm of one plus the variance of daily stock return) in Model (3). Refer to Appendix 1 for detailed variable definitions. All continuous variables are winsorized at the upper and lower 1% of the sample distribution. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A: The moderating effect of human capital intensity				
	(1)	(2)	(3)	(4)
	Ab_Abnormal_NetHire			
CEO tenure	−0.0004*** (−3.20)	−0.0003** (−2.18)	−0.0001 (−0.37)	−0.0002* (−1.67)
Human capital intensity	0.0055*** (2.63)			
CEO tenure*Human capital intensity	−0.0005** (−2.47)			
Human capital intensive industries		0.0133*** (2.99)		
CEO tenure*Human capital intensive industries		−0.0005** (−2.43)		
Labor skill			0.0125 (1.35)	
CEO tenure*Labor skill			−0.0011** (−2.00)	
Knowledge capital				−0.0009* (−1.66)
CEO tenure*Knowledge capital				−0.0001** (−2.23)
Control variables	Yes	Yes	Yes	Yes
Constant	0.0730*** (2.98)	0.0717*** (2.94)	0.0550** (2.51)	0.0725*** (2.97)
Observations	21,603	21,603	14,952	21,603
R-squared	0.172	0.172	0.168	0.171
Industry effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Firm clustered standard error	Yes	Yes	Yes	Yes
Panel B: The moderating effect of operational uncertainty				
	(1)	(2)	(3)	
	Ab_Abnormal_NetHire			
CEO tenure	−0.0000 (−0.28)	−0.0001 (−0.27)	−0.0001 (−0.39)	
STD_CFO	0.1182*** (3.54)			
CEO tenure*STD_CFO	−0.0069*** (−2.83)			
Idiosyncratic risk		0.7286*** (4.68)		
CEO tenure*Idiosyncratic risk		−0.0174* (−1.84)		
Total risk				0.5679*** (4.30)
CEO tenure*Total risk				−0.0140* (−1.66)
Control variables	Yes	Yes	Yes	Yes
Constant	0.0694*** (2.81)	0.0479* (1.93)	0.0535** (2.17)	
Observations	21,603	19,702	19,702	
R-squared	0.171	0.173	0.173	
Industry effects	Yes	Yes	Yes	
Year fixed effects	Yes	Yes	Yes	
Firm clustered standard error	Yes	Yes	Yes	

the flexibility of increasing production when needed. Our results suggest that longer-tenured CEOs are more experienced in forecasting changes in demand due to changes in the economic environment, and thus reduce labor investment inefficiency by alleviating under-hiring to avoid labor under-investment and mitigating under-firing to avoid labor over-investment.¹⁸

4.3. Cross-sectional analyses

Three mechanisms can lead to the observed positive association between CEO tenure and labor investment efficiency, namely advanced firm-specific knowledge, relaxed monitoring, and changing CEO risk-taking incentives. In this subsection, we conduct cross-sectional analyses to identify the primary driving force of the improved labor investment efficiency over a CEO's time in office and report the results in Table 4.

4.3.1. Advanced firm-specific knowledge mechanism

Cao and Rees (2020) note that employees in human capital-intensive firms tend to be more educated and experienced and possess higher labor skills. Labor adjustment costs, such as firing, hiring, and training, are higher in human capital-intensive firms since such costs increase rapidly with labor expertise and skills (Button, 1990; Hamermesh and Pfann, 1996; Ochoa, 2013; Pfann and Verspagen, 1989; Cao and Rees, 2020). Furthermore, competition in high-skilled sectors often involves 'talent wars' that increase labor adjustment costs, subjecting human capital-intensive firms to higher labor market frictions (Chang and Jo, 2019; Dolfen, 2006). Consequently, it is more difficult for these firms to adjust their labor investment efficiently. If long-tenured CEOs make better labor investment decisions due to expertise built over time in office, we expect to observe a stronger negative association between CEO tenure and labor investment inefficiency in human capital-intensive firms. To test our prediction, we add human capital intensity proxies and their interactions with CEO tenure to our baseline regression. Panel A of Table 4 displays the results.

In Model (1), *Human capital intensity* is measured with the ratio of R&D expenditures to total sales (Ghaly et al., 2015). The coefficient on CEO tenure is negative and significant at the 1 % level, suggesting a longer CEO tenure mitigates labor investment inefficiency in less human capital-intensive firms. The coefficient on the interaction term, *CEO tenure*Human capital intensity*, is negative and significant at the 5% level, suggesting that the positive impact of CEO tenure on labor investment efficiency is more pronounced in high human capital-intensive firms. In Model (2), we follow prior studies to define a firm as being human capital-intensive if the firm belongs to telecommunications, high-tech, and healthcare industries (Ben-Nasr and Ghouma, 2018; Ertugrul, 2013; Ghaly et al., 2015). *Human capital-intensive industries* is an indicator variable that equals one for firms in industries with the SIC codes of 283, 357, 36, 384, 48, and 80 and zero otherwise. In Model (3), we use labor skill data (*Labor skill*) from Belo et al. (2017) to proxy for the degree of reliance on skilled labor. *Labor skill* is the industry average number of employees working in occupations with a JobZones index of 4 or 5. Firms depending more on skilled labors are classified as more labor-intensive. In Model (4), we measure labor skill with the flows of a firm's knowledge capital in intangibles (*Knowledge capital*) as in Ewens et al. (2023). This proxy capitalizes prior flows of R&D expenditures to overcome the bias in the levels of intangible investments reported on firm balance sheets. The results in Models (2)–(4) echo that in Model (1).

In addition, the demand for labor depends on various parameters such as prices, demand shocks, and lagged employment (Nickell, 1986). Uncertainty reduces corporate sales, profits, and firm-level capital investment significantly (Bulan, 2005). It is more challenging for firms with greater operational uncertainty to predict the optimal level of labor requirement accurately (Habib and Hasan, 2021). They may need to adjust labor investment to changing market conditions frequently, leading to unpredictable and unprogrammable outcomes and potentially more inefficient labor investments.

We measure the extent of uncertainty with the standard deviation of cash flows from operations over the past five years (*STD_CFO*), the logarithm of one plus the variance of residuals estimated from the Fama-French three-factor model (*Idiosyncratic risk*), and the logarithm of one plus the variance of daily stock return (*Total risk*). We add these variables and their interactions with CEO tenure to our baseline regression and present the results in Models (1), (2), and (3) of Panel B in Table 4, respectively. The positive and significant coefficients on the three uncertainty variables indicate that firms with more operational uncertainty have lower labor investment efficiency. More importantly, the coefficients on the interaction terms between uncertainty and CEO tenure are negative and significant, suggesting that the positive effect of uncertainty on labor investment inefficiency is alleviated in firms with longer-tenured CEOs.

4.3.2. Relaxed monitoring mechanism

As monitors become more familiar with a CEO's ability and management style over the CEO's tenure, they may relax their monitoring, which could promote long-tenured CEOs to use their knowledge to increase labor investment efficiency. We measure governance quality with two external (the presence of a staggered board and the E-index) and two internal (the proportion of independent directors on the board and board size) governance variables and revisit our baseline model, adding these proxies and their interactions with CEO tenure.¹⁹ The coefficients of the interaction terms are all insignificant, inconsistent with the relaxed monitoring explanation. The results are not tabulated but available upon request.

¹⁸ Similarly, Ee et al. (2022) find that stock liquidity only reduces over-investment in labor through reducing over-hiring. Ghaly et al. (2020) show that long-term investors reduce labor investment inefficiency by mitigating under-hiring, under-firing, and over-hiring, but have no impact on over-firing.

¹⁹ We employ the staggered board classification used in Guernsey et al. (2022). We thank professor Serfling for providing the data at https://sites.google.com/utk.edu/matthew-serfling/publications_1.

4.3.3. Changing risk-taking incentives

Longer-tenured CEOs are subject to less career concerns than those with a shorter-tenure. The evolving career concerns over the tenure of a CEO could affect the CEO's risk-taking incentives, potentially resulting in more efficient labor investment in firms led by longer-tenured CEOs. CEO delta (the sensitivity of managerial wealth to stock price) exposes CEOs to more risk compare to diversified shareholders, enhancing their risk aversion (Holmstrom and Milgrom, 1991; Coles et al., 2006; Benson et al., 2019). CEO vega (the sensitivity of managerial wealth to stock return volatility) ties CEO wealth to stock return volatility and encourages CEOs to undertake risky investments (Guay, 1999; Coles et al., 2006; Benson et al., 2019). We revisit our baseline model by adding these CEO risk-taking incentive proxies and their interactions with *CEO tenure* and find that the coefficients of the interaction terms are insignificant. These results are inconsistent with the changing risk-taking incentives explanation. The results are not tabulated but available upon request.

4.4. CEO tenure, labor investment efficiency, and firm performance

We use mediation analysis as in Baron and Kenny (1986) to examine whether increased labor investment efficiency in firms with longer-tenured CEOs leads to better firm performance. Prior research has employed mediation analysis to identify the underlying channels through which variables of interest affect various corporate outcomes (Chen et al., 2019; Fu et al., 2020; Francis et al., 2021; James, 2023). Specifically, we use the structural equation modeling (SEM), in which the first regression regresses abnormal labor investment (*Ab_Abnormal_NetHire*) on *CEO tenure* and the second regression regresses firm performance (*ROA*) on *Ab_Abnormal_NetHire* and *CEO tenure*. We control *CEO age* and several firm characteristics that are potentially related to firm performance, including market-to-book value of common equity (*MTB*), firm size (*Size*), leverage (*Leverage*), dividend payout status (*DivDum*), and fixed assets investments (*Tangibles*).

The results are presented in Table 5. The negative and significant coefficient of *CEO tenure* in Model (2) shows that seasoned CEOs reduce labor investment inefficiency, consistent with our baseline results. Model (1) shows the standalone effects of *Ab_Abnormal_NetHire* and *CEO tenure* on *ROA*. As expected, labor investment inefficiency results in lower operating performance. The product of the coefficient of *Ab_Abnormal_NetHire* in Model (1) and the coefficient of *CEO tenure* in Model (2) captures the mediating effect of CEO tenure on firm performance through increasing labor investment efficiency. The Sobel test (Sobel, 1982) shows that the mediation effect is significant at a 5% level. These results indicate that longer-tenured CEOs adopt more efficient labor investment policies, which in turn enhances firm operating performance.

Table 5

CEO tenure, labor investment efficiency, and firm performance. This table presents the results of the tests of the effect of CEO tenure on firm performance through increased labor investment efficiency. Firm performance is measured with *ROA*. Model (2) shows the results of the first regression in the structural equation modeling (SEM), where abnormal labor investment (*Ab_Abnormal_NetHire*) is regressed on *CEO tenure* and the control variables. Model (1) shows the results of the second regression, where *ROA* is regressed on abnormal labor investment, CEO tenure, and the control variables. We control industry and year effects, where the industry is defined using the Fama and French 48 industry classification. Standard errors are clustered at the firm level. Refer to Appendix 1 for detailed variable definitions. All continuous variables are winsorized at the upper and lower 1% of the sample distribution. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	(1) ROA	(2) Ab_Abnormal_NetHire
Ab_Abnormal_NetHire	-0.0421*** (-7.95)	
CEO tenure	0.0008*** (8.34)	-0.0003** (-2.12)
CEO age	-0.0000 (-0.21)	-0.0006*** (-4.29)
MTB	0.0030*** (16.45)	0.0002 (0.69)
Size	0.0183*** (37.24)	-0.0057*** (-9.11)
Leverage	-0.0973*** (-23.58)	0.0127** (2.40)
DivDum	0.0152*** (9.73)	-0.0201*** (-10.04)
Tangible	-0.0234*** (-5.05)	-0.0388*** (-6.53)
Constant	-0.0674*** (-2.91)	0.1842*** (6.20)
Sobel test for the significance of the mediation effect	0.02	
P-value		
Observations	21,603	21,603
Industry effects	Yes	Yes
Year fixed effects	Yes	Yes
Firm clustered standard error	Yes	Yes

Table 6

Entropy-balanced sample. This table presents the test results using the Entropy-balancing approach. Panel A presents the distribution comparison of the control variables after entropy balancing, conducted by reweighting the control variables in the baseline regression to match the mean, variance, and skewness between firms with long-tenured CEOs (CEO tenure above the sample median) and those with short-tenured CEOs (CEO tenure below the sample median). *Std.Diff.* is the differences in the mean values of the control variables between the treated (firms with long-tenured CEOs) and the control samples (those with short-tenured CEOs) standardized by the standard deviation of the treated sample. *Var.Ratio* is the ratio of the variance of the control variables in the treated sample scaled by the variance of the control variables in the control sample. Panel B reports the regression results using the entropy-balanced sample. All independent variables are lagged one year from the dependent variable. We control industry and year effects, where the industry is defined using the Fama and French 48 industry classification. Standard errors are clustered at the firm level. Refer to Appendix 1 for detailed variable definitions. All continuous variables are winsorized at the upper and lower 1% of the sample distribution. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Distribution comparison post entropy balancing								
	CEO tenure > Median			CEO tenure < Median			<i>Std.Diff.</i>	<i>Var.Ratio</i>
	Mean	Variance	Skewness	Mean	Variance	Skewness		
CEO age	58.2200	57.3700	0.3234	58.2200	57.3700	0.3233	0.0000	1.0000
MTB	3.3910	14.6200	2.7930	3.3910	14.6200	2.7930	0.0000	1.0000
Size	7.4160	2.4470	0.3632	7.4160	2.4470	0.3632	0.0000	1.0000
Quick	1.8190	2.9170	2.3330	1.8190	2.9170	2.3330	0.0000	1.0000
Leverage	0.2063	0.0330	0.7356	0.2063	0.0330	0.7356	0.0000	1.0000
DivDum	0.5164	0.2498	−0.0656	0.5164	0.2498	−0.0656	0.0000	1.0000
STD_CFO	0.0517	0.0016	1.9140	0.0517	0.0016	1.9140	0.0000	1.0000
STD_Sale	0.2299	0.0425	1.9580	0.2299	0.0425	1.9580	0.0000	1.0000
Tangibles	0.2762	0.0524	1.0060	0.2762	0.0524	1.0060	0.0000	1.0000
Loss	0.1386	0.1194	2.0910	0.1386	0.1194	2.0910	0.0000	1.0000
Insti	0.7341	0.0434	−0.5082	0.7341	0.0434	−0.5082	0.0000	1.0000
STD_NetHire	0.1609	0.0331	3.2200	0.1609	0.0331	3.2200	0.0000	1.0000
Labor_Intensity	0.0059	0.0001	2.9140	0.0059	0.0001	2.9140	0.0000	1.0000
Union	0.0996	0.0074	1.8300	0.0996	0.0074	1.8300	0.0000	1.0000
AB_Abnormal_InvestOther	0.0898	0.0084	3.3010	0.0898	0.0084	3.3010	0.0000	1.0000

Panel B: Weighted regression using the entropy-balanced sample	
	(1)
	Ab_Abnormal_NetHire
CEO tenure	−0.0004** (−2.44)
CEO age	−0.0004* (−1.95)
MTB	−0.0003 (−1.01)
Size	−0.0035*** (−4.71)
Quick	0.0070*** (6.84)
Leverage	0.0055 (0.77)
DivDum	−0.0018 (−0.71)
STD_CFO	0.0974** (2.13)
STD_Sale	0.0315*** (3.76)
Tangible	−0.0115 (−1.20)
Loss	0.0074** (2.11)
Insti	0.0027 (0.42)
STD_NetHire	0.0583*** (7.43)
Labor_Intensity	−0.6129*** (−3.09)
Union	0.0668** (2.30)
AB_Abnormal_InvestOther	0.4161*** (18.82)
Constant	0.0564** (2.48)
Observations	21,603

(continued on next page)

Table 6 (continued)

Panel B: Weighted regression using the entropy-balanced sample	
	(1)
	Ab_Abnormal_NetHire
R-squared	0.158
Industry effects	Yes
Year fixed effects	Yes
Firm clustered standard error	Yes

Table 7

CEO turnover analysis. This table presents the results from regressing the change in labor investment inefficiency on the change in CEO tenure due to CEO turnover. We identify 688 CEO turnover events in our baseline sample. We require firms to have a minimum of two firm-year observations prior to and post CEO turnover to be included in the turnover analysis, resulting in 350 cases of turnover. Changes in labor investment inefficiency/CEO tenure are measured as the differences in the average values of labor investment inefficiency (*Ab_Abnormal_NetHire*)/the tenure of CEOs (*CEO tenure*) post and prior to CEO turnover, excluding the transition years. Models (1), (2), and (3) show the results from the 7-year, 5-year, and 3-year windows, respectively. We control industry and year effects, where the industry is defined using the Fama and French 48 industry classification. Standard errors are clustered at the firm and year levels. Refer to Appendix 1 for detailed variable definitions. All continuous variables are winsorized at the upper and lower 1% of the sample distribution. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)
	Dependent Variable = $\Delta Ab_Abnormal_NetHire$		
	7-year window	5-year window	3-year window
ΔCEO tenure	−0.0014*** (−2.59)	−0.0013** (−2.55)	−0.0016*** (−2.66)
CEO age	0.0004 (1.17)	0.0004 (1.28)	0.0005 (1.22)
MTB	0.0007 (0.83)	0.0010 (1.00)	0.0000 (0.03)
Size	−0.0068** (−2.17)	−0.0082** (−2.32)	−0.0096** (−2.11)
Quick	−0.0019 (−0.52)	−0.0025 (−0.61)	−0.0018 (−0.43)
Leverage	0.0275 (1.01)	0.0302 (1.01)	0.0449 (1.23)
DivDum	0.0188** (2.16)	0.0220** (2.34)	0.0141 (1.28)
STD_CFO	0.0854 (0.67)	0.0562 (0.35)	0.1074 (0.59)
STD_Sale	−0.0402** (−2.10)	−0.0436* (−1.96)	−0.0453 (−1.50)
Tangible	−0.0156 (−0.54)	−0.0106 (−0.35)	−0.0130 (−0.36)
Loss	0.0012 (0.19)	0.0036 (0.50)	0.0059 (0.60)
Insti	0.0170 (0.85)	0.0210 (0.97)	0.0081 (0.29)
STD_NetHire	−0.0546* (−1.68)	−0.0639* (−1.71)	−0.1109** (−2.43)
Labor_Intensity	−0.1077 (−0.13)	−0.1564 (−0.18)	0.6464 (0.71)
Union	−0.0638 (−0.91)	−0.0714 (−0.94)	−0.1653* (−1.91)
AB_Abnormal_InvestOther	0.0370 (1.25)	0.0554* (1.69)	0.0497 (1.00)
Constant	0.0276 (0.45)	0.0680 (1.18)	0.0678 (1.12)
Observations	3,861	3,202	2,226
R-squared	0.193	0.204	0.243
Industry effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Firm clustered standard error	Yes	Yes	Yes

5. Addressing endogeneity concerns

5.1. Entropy balancing approach

Systematic differences in firms with long-tenured CEOs and those with short-tenured ones could lead to spurious relation between CEO tenure and labor investment inefficiency. We employ the Entropy balancing (EB) approach to address this issue. The EB procedure is similar to, but more advantageous than, the propensity score matching (PSM). This approach minimizes the differences between

Table 8

Instrumental variable approach. This table presents the results of instrumental variable regression. In the first stage, we instrument *CEO tenure* with three variables, the median value of CEO tenure for firms in the same county as the focal firms (*CEOtenure_CT*), total number of CEOs for firms in the same county as the focal firms (*TotalCEO_CT*), and CEO tenure for the first year observation of firms included in our sample period (*FirstTenure*). In the second stage, we replace *CEO tenure* in our baseline with its predicted value (*Predicted CEO tenure*). We control industry and year effects, where the industry is defined using the Fama and French 48 industry classification. Standard errors are clustered at the firm and year levels. Refer to Appendix 1 for detailed variable definitions. All continuous variables are winsorized at the upper and lower 1% of the sample distribution. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)
	1st stage	2nd stage
	CEO tenure	Ab_Abnormal_NetHire
CEOtenure_CT	0.0320** (2.30)	
TotalCEO_CT	−0.1410*** (−3.10)	
FirstTenure	0.5177*** (97.44)	
Predicted CEO tenure		−0.0005** (−2.54)
CEO age	0.3677*** (68.29)	−0.0001 (−0.82)
MTB	0.0513*** (4.86)	−0.0004* (−1.67)
Size	−0.2162*** (−7.17)	−0.0037*** (−5.80)
Quick	0.2232*** (7.56)	0.0074*** (11.84)
Leverage	−1.3871*** (−5.63)	0.0079 (1.53)
DivDum	−0.3955*** (−4.26)	−0.0005 (−0.28)
STD_CFO	−2.7410** (−2.33)	0.0604** (2.45)
STD_Sale	1.8332*** (7.59)	0.0315*** (6.19)
Tangible	0.6655** (2.52)	0.0022 (0.40)
Loss	−0.7753*** (−6.85)	0.0069*** (2.91)
Insti	−0.5632** (−2.50)	0.0036 (0.76)
STD_NetHire	0.0802 (0.35)	0.0640*** (13.37)
Labor_Intensity	−20.6253*** (−2.99)	−0.5426*** (−3.75)
Union	−2.3559*** (−2.90)	0.0165 (0.97)
AB_Abnormal_InvestOther	−1.1897*** (−2.81)	0.4197*** (47.29)
Constant	−13.9123*** (−9.95)	0.0439 (1.52)
Observations	20,851	20,851
R-squared	0.485	0.160
Industry effects	Yes	Yes
Year fixed effects	Yes	Yes
Firm clustered standard error	Yes	Yes
Hansen J statistic		0.645
Hansen J statistic P-value		0.7243
Underidentification test (Kleibergen-Paap rk LM statistic):		1893.04
Weak identification test (Kleibergen-Paap rk Wald F statistic)		1043.349

firms with longer-tenured CEOs (the treated sample) and those with shorter-tenured CEOs (the control sample) by continuously adjusting the distributional moments such as means, variances, and skewness of the observations in the control sample, allowing less researchers' discretion than the PSM. In addition, the EB method retains all observations, making the results more generalized.

The treated and control samples include firms with *CEO tenure* above and below the sample median, respectively. Panel A of Table 6 displays the distribution comparison of the control variables after Entropy balancing. *Std.Diff.* is the differences in the mean values of the control variables between the treated and the control firms standardized by the standard deviation of the control variables in the treated sample. *Var.Ratio* is the ratio of the variance of the control variables in the treated sample scaled by the variance of the control variables in the control sample. *Stdz.Diff.* equals zero and *Var.Ratio* equals one across all variables, indicating that the treated and control firms are matched in their distributional moments. Panel B shows the regression result using the entropy-balanced sample. The result is qualitatively the same as those in the baseline regressions.

5.2. CEO turnover analysis

CEO tenure is relatively persistent, which may create endogeneity. We use CEO turnover as an exogenous shock to study the change in labor investment inefficiency. We classify firm-year observations with CEO turnover when the current CEO ID number and firm combination (Execucomp item coperol) is different from that in the previous year while the firm identifier remains the same. We identify 688 CEO turnover events in our sample. To mitigate the concern that the change in labor investment inefficiency around CEO turnover is due to noise, we require that all firms included in the analysis have a minimum of two observations prior to and post CEO turnover, resulting in 350 cases. Changes in labor investment inefficiency/CEO tenure are measured as the differences between the average values of *Ab_Abnormal_NetHire/CEO tenure* post and prior to CEO turnover over various windows, excluding the transition year. Models (1), (2), and (3) in Table 7 show the results from seven-year, five-year, and three-year windows, respectively. The coefficient on $\Delta CEO tenure$ is negative and significant across all three models, suggesting a negative association between increases in CEO tenure and labor investment inefficiency.

5.3. Instrumental variable approach

We employ the instrumental variable approach to further address the endogeneity concern arising from simultaneity and reverse causality (Wooldridge, 2002) and report the results in Table 8. We instrument *CEO tenure* with three variables: the median value of *CEO tenure* for firms in the same county as the focal firm excluding the focal firm (*CEOtenure_CT*), the total number of CEOs for firms in the same county as the focal firm excluding the focal firm (*TotalCEO_CT*), and *CEO tenure* of the first-year observation of firms in our sample period (*FirstTenure*). A focal firm's CEO tenure is expected to be positively related to CEO tenure in nearby firms. The segmented CEO labor market makes CEOs in surrounding firms potential successors of the focal firm's CEO. Therefore, the total number of CEOs around the focal firm is expected to be negatively related to its CEO's tenure. Lastly, the longer a CEO enters our sample period, the longer the CEO tenure in the later sample years. In addition, these variables are not expected to be directly related to the focal firm's labor investment efficiency. Collectively, these instruments satisfy both the exclusion and relevance criteria. Model (1) presents the first-stage result. All three instruments are significantly associated with *CEO tenure* in ways consistent with our expectations. The second-stage result in Model (2) shows that the predicted value of CEO tenure obtained from the first-stage regression (*Predicted CEO tenure*) is negative and significantly related to labor investment inefficiency. We conclude that our baseline results are likely to be causal.

5.4. The effects of other investments

CEO tenure may indirectly affect labor investments through its influence on other non-labor investments. If a firm simultaneously increases its capital and labor investments to facilitate its growth, the association between CEO tenure and labor investment efficiency can be driven by capital investments. Though we control for the effect of other abnormal non-labor investments (*AB_Abnormal_InvestOther*), the baseline regressions may not fully rule out the possibility that our results are driven by other contemporaneous non-labor investments. We use subsamples based on the association between labor and non-labor investment to further address this issue.

Following Ben-Nasr and Alshwer (2016), Jung et al. (2014), and Cao and Rees (2020), we examine the impact of four types of non-labor investments, including capital, R&D, advertising, and acquisition expenditures. The results are presented in Panels A, B, C, and D of Table 9, respectively. For each type of non-labor investment, we divide our sample into three subsamples, POSITIVE, NEGATIVE, and ZERO OR MISSING, based on the relation between net hiring and the specific non-labor investments. The POSITIVE subsample includes observations for which an increase (a decrease) in labor investments is accompanied by an increase (a decrease) in non-labor investments (i.e., a positive relation between labor and non-labor investments); the NEGATIVE subsample includes observations for which an increase (a decrease) in labor investments is accompanied by a decrease (an increase) in non-labor investments (i.e., a negative relation between labor and non-labor investments); and the ZERO OR MISSING subsample includes observations with missing values for non-labor investments.

If our baseline results are driven by other contemporaneous non-labor investments, we would expect that the baseline results be

Table 9

Effects from other non-labor investments. This table presents the results for the impact of non-labor investments on the CEO tenure-labor investment efficiency relation. Panel A/B/C/D displays the results for the subsample based on Capital/Research & development/Advertising /Acquisition expenditures. In all panels, Model (1) reports the results for the subsample of firms for which an increase (a decrease) in labor investment is accompanied with an increase (a decrease) in non-labor investment; Model (2) reports the results for the subsample of firms for which an increase (a decrease) in labor investment is accompanied with a decrease (an increase) in non-labor investment, and Model (3) reports the results for the subsample of firms with missing values of non-labor investments (i.e., firms without non-labor investments). We control industry and year effects, where the industry is defined using the Fama and French 48 industry classification. Standard errors are clustered at the firm level. Refer to Appendix 1 for detailed variable definitions. All continuous variables are winsorized at the upper and lower 1% of the sample distribution. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Capital expenditures				Panel B: Research & development		
	(1)	(2)	(3)	(1)	(2)	(3)
	Ab_Abnormal_NetHire			Ab_Abnormal_NetHire		
	Positive	Negative	Zero or Missing	Positive	Negative	Zero or Missing
CEO tenure	−0.0004** (−2.28)	−0.0004*** (−3.06)	0.0023 (1.16)	−0.0002* (−1.72)	−0.0010*** (−5.10)	−0.0003 (−1.47)
CEO age	−0.0002 (−0.96)	−0.0002 (−1.25)	0.0001 (0.05)	−0.0004 (−1.34)	0.0002 (1.09)	−0.0002 (−1.00)
MTB	−0.0002 (−0.70)	−0.0007** (−2.34)	−0.0057 (−0.81)	−0.0000 (−0.10)	−0.0006 (−1.45)	−0.0008** (−2.41)
Size	−0.0028*** (−3.38)	−0.0033*** (−4.79)	0.0124 (1.00)	−0.0014 (−1.22)	−0.0046*** (−4.40)	−0.0035*** (−3.99)
Quick	0.0058*** (5.96)	0.0077*** (7.82)	0.0212* (1.83)	0.0084*** (7.45)	0.0052*** (4.93)	0.0072*** (4.60)
Leverage	0.0014 (0.19)	0.0105 (1.56)	−0.0901 (−1.43)	−0.0010 (−0.09)	0.0268** (2.36)	−0.0031 (−0.43)
DivDum	−0.0059** (−2.25)	0.0019 (0.78)	−0.0356 (−1.15)	−0.0029 (−0.73)	0.0056* (1.68)	−0.0042 (−1.64)
STD_CFO	0.0984*** (2.59)	0.0458 (1.32)	0.7177* (1.80)	0.0946* (1.67)	0.1072** (2.29)	0.0313 (0.79)
STD_Sale	0.0555*** (6.45)	0.0020 (0.29)	−0.1287** (−2.58)	0.0733*** (4.85)	−0.0061 (−0.41)	0.0214*** (3.15)
Tangible	0.0019 (0.22)	0.0031 (0.46)	0.1515* (1.96)	−0.0118 (−0.76)	−0.0153 (−1.19)	0.0042 (0.57)
Loss	−0.0061* (−1.94)	0.0220*** (6.63)	0.0301 (0.98)	−0.0034 (−0.74)	0.0273*** (6.40)	0.0027 (0.75)
Insti	0.0130** (1.99)	−0.0121* (−1.95)	0.0239 (0.33)	0.0152 (1.49)	−0.0335*** (−3.68)	0.0125* (1.87)
STD_NetHire	0.0556*** (6.44)	0.0698*** (9.55)	0.1083* (1.68)	0.0483*** (3.23)	0.0689*** (6.09)	0.0614*** (7.33)
Labor_Intensity	−0.6798*** (−3.68)	−0.2197 (−1.33)	−3.1234* (−1.81)	−1.6941** (−2.31)	−0.1355 (−0.22)	−0.4625*** (−3.01)
Union	0.0325 (1.35)	−0.0113 (−0.52)	−0.1205 (−0.60)	−0.0255 (−0.57)	−0.0403 (−0.91)	0.0219 (1.01)
AB_Abnormal_InvestOther	0.4028*** (19.89)	0.3400*** (15.58)	0.2378 (1.39)	0.4207*** (17.74)	0.2770*** (10.10)	0.3887*** (15.14)
Constant	0.0873** (2.49)	0.0382** (2.48)	−0.0854 (−0.64)	0.0723** (2.14)	0.0943* (1.88)	0.0733** (2.20)
Observations	10,640	10,910	46	5,502	5,489	10,613
R-squared	0.197	0.171	0.653	0.257	0.181	0.146
Industry effects	Yes	Yes	No	Yes	Yes	Yes
Year fixed effects	Yes	Yes	No	Yes	Yes	Yes
Firm clustered standard error	Yes	Yes	Yes	Yes	Yes	Yes

Panel C: Advertising expenditures			Panel D: Acquisition expenditures			
	(1)	(2)	(3)	(1)	(2)	(3)
	Ab_Abnormal_NetHire			Ab_Abnormal_NetHire		
	Positive	Negative	Zero or Missing	Positive	Negative	Zero or Missing
CEO tenure	−0.0004* (−1.65)	−0.0006*** (−2.87)	−0.0004** (−2.57)	−0.0005** (−2.36)	−0.0004* (−1.98)	−0.0005*** (−2.80)
CEO age	−0.0000 (−0.03)	−0.0001 (−0.44)	−0.0002 (−1.43)	−0.0002 (−0.95)	−0.0001 (−0.43)	−0.0002 (−1.34)
MTB	−0.0001 (−0.16)	−0.0002 (−0.36)	−0.0008** (−2.37)	0.0003 (0.67)	−0.0003 (−0.82)	−0.0007*** (−2.60)
Size	−0.0009 (−0.82)	−0.0051*** (−4.45)	−0.0034*** (−4.30)	−0.0034*** (−3.36)	−0.0042*** (−4.38)	−0.0027*** (−3.33)
Quick	0.0071*** (4.73)	0.0046*** (3.26)	0.0074*** (8.17)	0.0054*** (4.01)	0.0090*** (5.92)	0.0074*** (8.22)
Leverage	0.0044 (0.46)	0.0107 (1.00)	0.0056 (0.76)	−0.0217** (−2.49)	0.0244*** (2.94)	0.0098 (1.29)

(continued on next page)

Table 9 (continued)

DivDum	−0.0098** (−2.43)	0.0026 (0.71)	0.0003 (0.11)	−0.0007 (−0.22)	−0.0042 (−1.45)	−0.0008 (−0.32)
STD_CFO	0.1196** (2.17)	0.0975* (1.85)	0.0630* (1.70)	0.0433 (0.87)	0.0834* (1.72)	0.1018*** (2.78)
STD_Sale	0.0669*** (5.15)	−0.0110 (−0.98)	0.0257*** (3.49)	0.0321*** (3.25)	0.0119 (1.19)	0.0283*** (3.53)
Tangible	−0.0022 (−0.17)	0.0079 (0.66)	−0.0041 (−0.57)	−0.0001 (−0.01)	0.0115 (1.11)	0.0010 (0.13)
Loss	−0.0094** (−2.03)	0.0071 (1.34)	0.0124*** (4.03)	0.0037 (0.87)	0.0129*** (3.08)	0.0067** (2.07)
Insti	0.0107 (1.17)	−0.0079 (−0.85)	−0.0006 (−0.08)	0.0070 (0.79)	−0.0042 (−0.50)	0.0001 (0.02)
STD_NetHire	0.0489*** (3.43)	0.0620*** (4.95)	0.0619*** (7.89)	0.0319*** (3.39)	0.0781*** (7.84)	0.0543*** (5.73)
Labor_Intensity	−0.2653 (−1.08)	−0.3814 (−1.61)	−0.6308*** (−3.30)	−0.7079*** (−3.21)	−0.5179** (−2.16)	−0.2099 (−1.09)
Union	−0.0149 (−0.43)	0.0045 (0.13)	0.0274 (1.15)	0.0518 (1.47)	−0.0125 (−0.43)	−0.0039 (−0.17)
AB_Abnormal_InvestOther	0.3680*** (10.79)	0.4089*** (11.33)	0.3736*** (19.47)	0.4070*** (14.63)	0.4104*** (16.53)	0.3226*** (14.53)
Constant	−0.0435* (−1.72)	0.0541** (2.34)	0.0911*** (2.67)	0.0411 (1.51)	0.0628 (1.39)	0.0942*** (2.80)
Observations	4,036	4,005	13,496	6,523	6,949	10,429
R-squared	0.202	0.188	0.172	0.174	0.202	0.158
Industry effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm clustered standard error	Yes	Yes	Yes	Yes	Yes	Yes

observed in the POSITIVE subsamples only. However, the estimated coefficient on *CEO tenure* remains negative and significant across most models, suggesting that non-labor investments do not confound our conclusion.²⁰

5.5. Controlling for information asymmetry

Prior research recognizes that information asymmetry affects corporate investment decisions (Hubbard, 1998; Stein, 2003) and labor investment efficiency (Jung et al., 2014; Ben-Nasr and Alshwer, 2016; Lee and Mo, 2020; Sualihu et al., 2021b; Ee et al., 2022). Since CEO tenure can be associated with information asymmetry through its impact on corporate governance, we control for the level of information asymmetry to isolate the impact of CEO tenure on labor investment efficiency.²¹ The literature has identified that the number of analysts following a firm (Hong et al., 2000; Chang et al., 2006; Bowen et al., 2008), bid-ask spread (Glosten and Milgrom, 1985; Easley and O'hara, 1987; Glosten and Harris, 1988; Kim and Verrecchia, 1994; James et al., 2021), idiosyncratic volatility (Bhagat et al., 1985; Blackwell et al., 1990; Kelly, 2005; James et al., 2021), stock return synchronicity (Dasgupta et al., 2010; Aslan et al., 2011; Maffett, 2012; James et al., 2021), and trading volume (Lof and Bommel, 2023) affect information asymmetry. We, therefore, proxy the level of information asymmetry with the number of analysts following a firm (*Analysts*), bid-ask spread (*Bid-ask spread*), idiosyncratic volatility (*Idiorisk_4Factor* and *Idiorisk_MM* estimated using the Carhart four-factor and market models, respectively), the extent of stock return synchronicity (*R2_4Factor* and *R2_MM* estimated using the Carhart four-factor and market models, respectively), trading volume (*Vol*), and the coefficient of variation of trading volume (*VCV*).²² Higher values of *Analysts*, *R2_4Factor*, and *R2_MM* and lower values of *Idiorisk_4Factor*, *Idiorisk_MM*, *Vol*, *VCV*, and *Bid-ask spread* suggest lower information asymmetry. Financial analyst data are obtained from I/B/E/S, and data used to construct other information asymmetry variables are from CRSP. Refer to Appendix 1 for definitions and constructions of these variables.

We add the proxies for information asymmetry to our baseline model and rerun the analysis. Table 10 displays the results. As expected, lower information asymmetry is associated with higher labor investment efficiency, evidenced by negative and significant coefficients of *Analysts*, *R2_4Factor*, and *R2_MM* and positive and significant coefficients of *Idiorisk_4Factor*, *Idiorisk_MM*, *Vol*, *VCV*, and *Bid-ask spread*.²³ More importantly, the coefficient of *CEO tenure* is negative and significant at 1 % in all models, indicating that the impact of CEO tenure on labor investment efficiency is above and beyond the impact of information asymmetry.

5.6. Additional analyses

To further validate our findings, we conduct additional analysis and present the results in Appendix 3. Our first set of analyses uses three alternative measures of labor investment inefficiency. First, we follow Cella (2020) and Ben-Nasr and Alshwer (2016) to measure

²⁰ The coefficient on *CEO tenure* loses the significance in only two of the ZERO OR MISSING subsamples in Panels A and B.

²¹ We thank an anonymous reviewer for suggesting this analysis.

²² The Carhart four-factor model is developed on the Fama-French three-factor model (Fama and French, 1993; Carhart, 1997).

²³ The result is consistent with the literature, such as Lee and Mo (2020).

Table 10

Controlling for information asymmetry. This table reports regression results of the effect of CEO tenure on labor investment efficiency while controlling for the impact of information asymmetry. The level of information asymmetry is proxied with the number of analysts following a firm (*Analysts*), bid-ask spread (*Bid-ask spread*), idiosyncratic volatility (*Idiorisk_4Factor* and *Idiorisk_MM* estimated using the Carhart four-factor and market models, respectively), the extent of stock return synchronicity (*R2_4Factor* and *R2_MM* estimated using the Carhart four-factor and market models, respectively), trading volume (*Vol*), and the coefficient of variation of trading volume (*VCV*) in Models 1–8, respectively. We control industry and year-fixed effects, where the industry is defined using the Fama and French 48 industry classification. Standard errors are clustered at the firm level. Refer to Appendix 1 for detailed variable definitions. All continuous variables are winsorized at the upper and lower 1% of the sample distribution. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Ab_Abnormal_NetHire							
CEO tenure	−0.0005*** (−3.80)	−0.0004*** (−3.25)	−0.0004*** (−3.26)	−0.0004*** (−3.24)	−0.0004*** (−3.25)	−0.0004*** (−3.40)	−0.0005*** (−3.61)	−0.0005*** (−3.35)
CEO age	−0.0001 (−1.10)	−0.0001 (−1.08)	−0.0001 (−1.07)	−0.0002 (−1.14)	−0.0002 (−1.14)	−0.0001 (−0.69)	−0.0001 (−0.77)	−0.0001 (−0.65)
Analysts	−0.0045** (−2.47)							
Idiorisk_4Factor		3.1301*** (2.69)						
Idiorisk_MM			3.0878*** (2.79)					
R2_4Factor				−0.0195** (−2.28)				
R2_MM					−0.0203** (−2.33)			
Vol						0.0037*** (3.38)		
VCV							0.0110*** (4.35)	
Bid-ask spread								0.4354** (2.05)
MTB	−0.0005** (−2.08)	−0.0006*** (−2.67)	−0.0006*** (−2.67)	−0.0006*** (−2.63)	−0.0006*** (−2.63)	−0.0004* (−1.82)	−0.0005** (−2.29)	−0.0005* (−1.66)
Size	−0.0021** (−2.36)	−0.0029*** (−4.58)	−0.0029*** (−4.56)	−0.0028*** (−4.26)	−0.0027*** (−4.10)	−0.0062*** (−5.87)	−0.0020*** (−2.98)	−0.0029*** (−3.89)
Quick	0.0072*** (9.34)	0.0070*** (9.43)	0.0070*** (9.42)	0.0070*** (9.49)	0.0070*** (9.47)	0.0071*** (9.23)	0.0072*** (9.34)	0.0075*** (9.07)
Leverage	0.0041 (0.75)	0.0046 (0.82)	0.0046 (0.82)	0.0051 (0.91)	0.0053 (0.95)	0.0039 (0.69)	0.0046 (0.83)	0.0085 (1.33)
DivDum	−0.0008 (−0.43)	−0.0011 (−0.58)	−0.0011 (−0.56)	−0.0015 (−0.75)	−0.0014 (−0.74)	0.0001 (0.07)	−0.0006 (−0.30)	−0.0000 (−0.00)
STD_CFO	0.0672** (2.37)	0.0563** (2.00)	0.0558** (1.98)	0.0608** (2.15)	0.0605** (2.14)	0.0435 (1.50)	0.0594** (2.07)	0.0472 (1.44)
STD_Sale	0.0285*** (4.84)	0.0245*** (4.12)	0.0245*** (4.12)	0.0251*** (4.22)	0.0251*** (4.22)	0.0270*** (4.47)	0.0270*** (4.45)	0.0320*** (4.54)
Tangible	0.0007 (0.11)	−0.0015 (−0.24)	−0.0016 (−0.25)	−0.0012 (−0.19)	−0.0015 (−0.24)	−0.0027 (−0.42)	−0.0004 (−0.06)	−0.0027 (−0.39)
Loss	0.0082*** (3.33)	0.0059** (2.35)	0.0058** (2.30)	0.0071*** (2.87)	0.0070*** (2.83)	0.0065** (2.55)	0.0085*** (3.38)	0.0072** (2.50)
Insti	0.0048 (0.91)	0.0055 (1.11)	0.0055 (1.10)	0.0045 (0.91)	0.0043 (0.86)	0.0023 (0.45)	0.006 (1.18)	0.0070 (1.22)
STD_NetHire	0.0594*** (9.39)	0.0608*** (9.24)	0.0607*** (9.23)	0.0613*** (9.33)	0.0612*** (9.32)	0.0584*** (9.24)	0.0595*** (9.46)	0.0635*** (8.76)

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Table 10 (continued)

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Ab_Abnormal_NetHire							
Labor_Intensity	−0.5696*** (−3.75)	−0.5192*** (−3.51)	−0.5179*** (−3.50)	−0.5321*** (−3.59)	−0.5320*** (−3.59)	−0.4942*** (−3.16)	−0.5243*** (−3.37)	−0.5211*** (−3.30)
Union	0.0061 (0.32)	0.0022 (0.11)	0.0023 (0.12)	−0.0011 (−0.05)	−0.0014 (−0.07)	0.0049 (0.25)	0.0031 (0.16)	0.0236 (1.10)
AB_Abnormal_InvestOther	0.3734*** (23.77)	0.3800*** (23.32)	0.3801*** (23.32)	0.3802*** (23.35)	0.3802*** (23.34)	0.3805*** (23.78)	0.3799*** (23.79)	0.4264*** (22.02)
Constant	0.0604** (2.22)	0.0599** (2.20)	0.0599** (2.20)	0.0618** (2.27)	0.0615** (2.26)	0.0364 (1.33)	0.0339 (1.25)	0.0502** (1.99)
Observations	20,743	19,704	19,704	19,704	19,704	19,704	19,704	19,500
R-squared	0.173	0.176	0.176	0.175	0.175	0.175	0.175	0.164
Industry effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm clustered standard error	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

unexpected net hiring as the absolute value of the difference between a firm's actual net hiring and the median level of net hiring at the industry-year level. The more a firm's net hiring deviates from its industry peers, the more inefficient its labor investment. Second, we estimate net hiring as the function of lagged sales growth rate as in Biddle et al. (2009) and Jung et al. (2014) and use the absolute value of the residual as a proxy for labor investment inefficiency. Lastly, we augment Equation (1) by controlling for capital investments, R&D expenditures, acquisitions, PPE (property, plant, and equipment balances), industry unionization levels, prior year's net hiring, and the logarithm of GDP per capita to estimate the expected net hiring (Jung et al., 2014). We re-run the baseline regression by replacing *Ab_Abnormal_NetHire* with these three alternative proxies for labor investment inefficiency. Panel A displays the results. The results are consistent with those in Table 3, suggesting our results are robust to alternative proxies for abnormal net hiring.

In addition, we add more control variables that are associated with either CEO tenure or labor investment efficiency to address the omitted variable concern. Specifically, we include the log of the number of CEOs in the same state to control for local CEO supply, the average CEO tenure of the CEOs in the same state to control for the impact of general CEO tenure in a state, and the average director tenure to control for the impact of board directors on CEO tenure. Panel B of Appendix 3 displays the results. The results continue to hold.

Lastly, newly hired CEOs are less experienced than those in the CEO position for years. Meanwhile, due to career concerns, they have stronger incentives to impress the market for better compensation and job security. They may engage in either too conservative or aggressive and risk-taking actions, which can result in inefficient labor investment. Unlike newly hired CEOs, CEOs near retirement may lack incentives to learn new skills or adapt to the changing environment, leading to inefficient labor investment. On the other hand, given that CEOs' retirement compensation mainly relies on their performance in the final years of service (Dechow and Sloan, 1991; Antia et al., 2010), CEOs near retirement may be more cautious in selecting labor investment policies, leading to more efficient labor investment. To rule out possible bias from newly hired CEOs and those reaching retirement age, we factor in the effects of CEOs on both ends of CEO tenure. *Tenure3* equals one for CEOs with a tenure of less than or equal to three years and zero otherwise. *Last* equals one for CEOs in their last year in the CEO position and zero otherwise (Serfling, 2014). We augment the baseline regression by adding the two indicator variables (Model (1) of Panel C) and the two indicator variables and their interactions with *CEO tenure* (Model (2) of Panel C). The results echo those in the baseline model, suggesting that newly hired CEOs or CEOs near retirement do not drive our baseline results.

6. Conclusion

As an important factor of production, human capital is vital to the success of all firms. Optimal employment decisions are critically important. We examine whether and how CEO tenure affects corporate labor investment. Using the deviation of labor investment from the level justified by a firm's economic fundamentals to measure labor investment inefficiency, we find that labor investment inefficiency decreases with CEO tenure and the effect is more pronounced in firms where managerial expertise is more essential to labor investment, such as those with high human capital intensity and greater operating uncertainty. We also document that the increased labor investment efficiency in firms with longer-tenured CEOs leads to better operating performance. The results are consistent with the expertise view of longer-tenured CEOs, suggesting that longer-tenured CEOs can serve as an important source of firm value.

Our research offers important practical value to various stakeholders. First, our findings would be useful to investors by helping them make more informed decisions about which firms to invest in. Knowing that longer CEO tenure improves labor market efficiency, they may become more willing to invest in firms with long-tenured CEOs. Second, our research emphasizes the relevance of long-term thinking and preparation in leadership strategies of CEOs. CEOs may be incentivized to remain in their firms for longer periods of time if they acknowledge the positive effect of accumulated firm-specific knowledge on economic outcomes. Moreover, our findings can benefit boards of directors in CEO succession planning. If longer CEO tenure is linked with greater labor market efficiency, boards may intend to select CEOs who are likely to remain with their firms for an extended time. Third, our research should help policymakers make better judgements in labor-related laws and regulations. Policymakers may be more inclined to support measures that encourage corporations to keep their CEOs for longer periods of time. Finally, corporate reformers may hesitate to advocate universally capping CEO tenure if they recognize the value of long-tenured CEOs on labor investment.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix 1:. Variable definitions

NetHire	The percentage change in the number of employees from time t-1 to time t
AB_Abnormal_NetHire	The absolute value of the difference between the actual level of labor investment and its predicted level using the following equation: $NetHire_{it+1} = \beta_0 + \beta_1 SGR_{it} + \beta_2 SGR_{it+1} + \beta_3 \Delta ROA_{it} + \beta_4 \Delta ROA_{it+1} +$

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NetHire	The percentage change in the number of employees from time t-1 to time t
	$\beta_5 ROA_{it+1} + \beta_6 Return_{it+1} + \beta_7 Size_R_{it} + \beta_8 Quick_{it} + \beta_9 \Delta Quick_{it} + \beta_{10} \Delta Quick_{it+1} + \beta_{11} Leverage_{it} + \beta_{12} LossBin1_{it} + \beta_{13} LossBin2_{it} + \beta_{14} LossBin3_{it} + \beta_{15} LossBin4_{it} + \beta_{16} LossBin5_{it} + \varepsilon_{it+1}$
CEO tenure	The number of years that an individual has served as the CEO
Ln (1 + CEO tenure)	The natural logarithm of one plus the number of years an individual has served as the CEO
CEO age	Age of the CEO
SGR	The percentage change in sales from time t-1 to time t
ROA	Net income scaled by lagged total assets
Δ ROA	Change in ROA
Return	Annualized monthly stock return during fiscal year
SIZE	The market value of equity of a firm (in million dollars)
Size	The natural logarithm of SIZE (i.e., a firm's market value of equity)
Size_R	Percentile rank of the logarithm of Size
Quick	The sum of cash and accounts receivables scaled by current liabilities
Δ Quick	Change in quick ratio
LossBin	There are five separate loss bins in each 0.005 interval for ROA from 0 to -0.025. Specifically, LossBin1 is equal to one if ROA is between -0.005 and 0, LossBin2 is equal to one if ROA is between -0.01 to -0.005, LossBin3 is equal to one if ROA is between -0.015 and -0.01, LossBin4 is equal to one if ROA is between -0.02 to -0.015, and LossBin5 is equal to one if ROA is between -0.025 to -0.02
Leverage	The sum of short-term and long-term debt scaled by total assets
MTB	Market-to-book value of equity, constructed as the product of the number of shares outstanding and the stock price at the fiscal year end scaled by the book value of shareholders' equity
DivDum	Indicator variable equal to one if a firm paid dividends in a fiscal year
STD_CFO	Standard deviation of firms' cash flows from operations over the past five years
STD_Sale	Standard deviation of firms' sales over the past five years
Tangibles	Property, plant, and equipment divided by total assets
Loss	Indicator variable equal to one if a firm has negative ROA in a fiscal year
Insti	Institutional ownership, constructed as common shares held by institutional owners scaled by total number of shares outstanding
STD_NetHire	Standard deviation of firms' changes in the number of employees over the past five years.
Labor_Intensity	The number of employees scaled by lagged total assets.
Union	Industry-level rate of labor unionization. The database is available at https://www.unionstats.com . Hirsch and Macpherson (2003) detail the construction of the database
AB_Abnormal_InvestOther	Abnormal non-labor investment, constructed as the absolute value of the residuals from the model regressing other investments on the lagged sales growth rate, where other investments include capital expenditures, acquisition expenditures, and research and development expenses less the proceeds from the sale of property, plant, and equipment.
Human capital intensity	The ratio of R&D expenditures to total sales
Human capital intensive industries	All subcategories of the telecommunications, high-tech, and healthcare industries with the two- and three-digit SIC codes of 283, 357, 36, 384, 48 and 80
Labor skill	Labor skills data from Belo et al. (2017) that uses the industry average number of employees working in occupations with a JobZones index equal to 4 or 5 as a proxy for the degree of reliance on skilled labor
Knowledge capital	The flows of a firm's knowledge capital in the intangibles defined in Ewens et al. (2023). The data is obtained from https://github.com/michaelebens/Intangible-capital-stocks .
Idiosyncratic risk	Logarithm of one plus the variance of residuals estimated from the Fama-French 3-factor model
Total risk	Logarithm of one plus the variance of daily stock return
Staggered board	Indicator variable that equals one for those with a classified board, and zero otherwise. We obtain the data from https://sites.google.com/utk.edu/matthew-serfling/publications_1
E-index	E-index is constructed as in Bebchuk et al. (2009). As in Bebchuk et al. (2009), we assume the index score is the same as that of the previous year when missing
CEO Delta	The dollar change in CEO equity portfolio wealth for a 1 % change in the stock price
CEO Vega	The dollar change in CEO equity portfolio wealth for a 1 % change in the annualized standard deviation of stock returns
Prop_Ind_Dir	The proportion of independent directors on the board
Bdsize	The natural logarithm of the number of directors on the board
Log(number of CEO in the same state)	The natural logarithm of the number of CEOs in the same state
Avg_CEO tenure (in the same state)	The average CEO tenure for firms headquartered in the same state
Avg_director Tenure	The average tenure of all directors on the board
Analysts	The logarithm of the number of analysts following a firm
Idiorisk_4Factor	The logarithm of one plus the variance of residuals estimated from the Fama-French and Carhart 4-factor model of daily returns: $R_{i,t} = \beta_{m,t} R_{m,t} + \beta_{smb,t} R_{smb,t} + \beta_{hml,t} R_{hml,t} + \beta_{umd,t} R_{umd,t} + \varepsilon_{i,t}$, where where $R_{m,t}$ is the daily market return, R_{smb} is the premium that small stocks earn over large stocks, R_{hml} represents the premium that stocks with a high book value to market value of assets earn over those with a low book value to market value of assets, and R_{umd} is the premium earned on a portfolio formed from a long position in past winners and a short position in past losers.
Idiorisk_MM	The logarithm of one plus the variance of residuals estimated from the market model: $R_{i,t} = \alpha_{i,t} + \beta_{i,t} R_{m,t} + \varepsilon_{i,t}$, where where $R_{i,t}$ is the daily stock return and $R_{m,t}$ is the daily market return.
R2_4Factor	The logarithm of one plus R^2 estimated using the Fama-French and Carhart 4-factor model.
R2_MM	The logarithm of one plus R^2 estimated using the market model
Vol	The logarithm of one plus the average daily trading volume
VCV	The ratio of the standard deviation of daily trading volume to its mean
Bid-ask spread	The logarithm of 1 plus the bid-ask spread where the bid-ask spread is the ratio of high-to-low prices a day as defined in Corwin and Schultz (2012), averaged to annual frequency to match with our data.

Appendix 2:. Correlations

This table reports the Pearson pairwise correlation matrix of the variables. Labor investment efficiency (*Ab_Abnormal_NetHire*) is the absolute value of the difference between the actual value of labor investment and the expected value estimated from Equation (1). *CEO tenure* is the number of years that an individual serving as the CEO. Refer to Appendix 1 for detailed variable definitions. All continuous variables are winsorized at the upper and lower 1 % of the sample distribution. Values in bold indicate statistical significance at the 5 % level or above.

	Ab_Abnormal_NetHire	CEO tenure	CEO age	MTB	Size	Quick	Leverage	DivDum
CEO tenure	−0.0247							
CEO age	−0.0651	0.4314						
MTB	0.0022	0.0157	−0.0367					
Size	−0.1202	−0.0645	0.0694	0.2591				
Quick	0.1424	0.1009	−0.0698	0.0305	−0.175			
Leverage	−0.0271	−0.0867	0.0688	−0.0286	0.1343	−0.3385		
DivDum	−0.1184	−0.0341	0.145	0.0054	0.3138	−0.271	0.117	
STD_CFO	0.1568	0.0276	−0.1163	0.1538	−0.2274	0.2763	−0.238	−0.3075
STD_Sale	0.1027	0.0518	−0.0709	0.064	−0.1536	−0.0146	−0.1766	−0.1882
Tangibles	−0.0457	−0.0448	0.0862	−0.1181	0.0811	−0.334	0.279	0.2255
Loss	0.0857	−0.0453	−0.0527	−0.0871	−0.2745	0.0767	0.0295	−0.222
Insti	−0.0396	−0.06	−0.0387	0.0437	0.2102	0.025	−0.0041	−0.1101
STD_NetHire	0.168	−0.0155	−0.0839	−0.0343	−0.1328	0.0751	0.0561	−0.2178
Labor_Intensity	−0.034	0.0211	0.0243	−0.019	−0.2216	−0.1728	−0.0978	−0.0141
Union	−0.0288	−0.0517	0.0606	−0.1183	0.0167	−0.1958	0.24	0.2238
AB_Abnormal_InvestOther	0.3452	−0.0063	−0.0477	0.0655	−0.073	0.0907	−0.011	−0.1027
		STD_Sale	Tangibles	Loss	Insti	(STD_NetHire	Labor_Intensity	Union
STD_Sale	0.3962							
Tangibles	−0.1677	−0.1376						
Loss	0.1489	−0.0213	−0.0188					
Insti	−0.0811	−0.0417	−0.1675	−0.0857				
STD_NetHire	0.1816	0.2794	−0.0868	0.1406	−0.0369			
Labor_Intensity	0.0211	0.2158	0.0866	−0.0367	−0.0672	−0.0456		
Union	−0.1684	−0.1038	0.4578	−0.0414	−0.2382	−0.0523	−0.1614	
AB_Abnormal_InvestOther	0.1585	0.0599	−0.0878	0.0562	0.0097	0.0709	−0.0424	−0.0669

Appendix 3:. Other robustness checks

This table presents the results from additional robustness checks. Panel A reports the results of using three alternative labor investment efficiency proxies. Model (1) shows the results from estimating abnormal net hiring as the deviation from the industry-year median; Model (2) shows the results from estimating net hiring using only lagged sale growth rate; and Model (3) shows the results from using extended [Pinnuck and Lillis \(2007\)](#) model to estimate normal hiring by augmenting it with controls for capital investments, R&D expenditures, acquisitions, property, plant, and equipment balances, industry unionization levels, and prior year's net hiring. Panel B reports the results by including more control variables to address the omitted variable bias issues. Models (1), (2), and (3) report the results by including the log of number of CEOs in the same state, the average CEO tenure in the same state, and the average director tenure, respectively. Panel C reports the results by controlling the effects of newly hired CEOs and CEOs near retirement. *Tenure 3* is an indicator variable that equals one for CEOs with a tenure less than or equal to three years, and zero otherwise. *Last* is an indicator variable that equals one for CEOs whose term ends in one year, and zero otherwise. All independent variables are lagged one year from the dependent variable. We control industry and year effects, where the industry is defined using Fama and French 48 industry classification. Standard errors are clustered at the firm level. Refer to Appendix 1 for detailed variable definitions. All continuous variables are winsorized at the upper and lower 1 % of the sample distribution. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

Panel A: Alternative proxies for labor investment inefficiency

	(1)	(2)	(3)
	Ab_Abnormal_NetHire		
CEO tenure	−0.0003** (−2.04)	−0.0004*** (−2.88)	−0.0004*** (−2.84)

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	(1)	(2)	(3)
	(23.86)	(23.99)	(21.22)
Control	Yes	Yes	Yes
Constant	0.0789*** (3.09)	0.0726*** (3.03)	0.0565** (2.21)
Observations	21,603	21,603	21,410
R-squared	0.192	0.175	0.136
Industry effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Firm clustered standard error	Yes	Yes	Yes

Panel B: Omitted variable bias

	(1)	(2)	(3)
	Ab_Abnormal_NetHire		
CEO tenure	−0.0004*** (−3.50)	−0.0004*** (−3.56)	−0.0003** (−2.13)
Log(number of CEOs in the same state)	−0.0007 (−0.78)		
Avg_CEO tenure (in the same state)		0.0004 (0.42)	
Avg_director Tenure			−0.0005* (−1.85)
Control variables	Yes (24.65)	Yes (24.64)	Yes (21.31)
Constant	0.0768*** (3.06)	0.0701*** (2.77)	0.0986*** (5.88)
Observations	21,603	21,603	15,108
R-squared	0.171	0.171	0.167
Industry effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Firm clustered standard error	Yes	Yes	Yes

Panel C: Newly hired CEOs and CEOs near retirement

	(1)	(2)
	Ab_Abnormal_NetHire	
CEO tenure	−0.0005*** (−3.33)	−0.0005*** (−3.18)
Tenure3	−0.0011 (−0.57)	−0.0015 (−0.57)
CEO tenure*Tenure3		0.0003 (0.25)
Last	0.0053* (1.79)	0.0060* (1.66)
CEO tenure*Last		−0.0001 (−0.39)
Control variables	Yes	Yes
Constant	0.0617** (2.36)	0.0617** (2.36)
Observations	21,603	21,603
R-squared	0.173	0.173
Industry effects	Yes	Yes
Year fixed effects	Yes	Yes
Firm clustered standard error	Yes	Yes

Data availability

All data used in this study are publicly available from the sources indicated in the paper.

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